

44
YEARS
OF EXCELLENCE

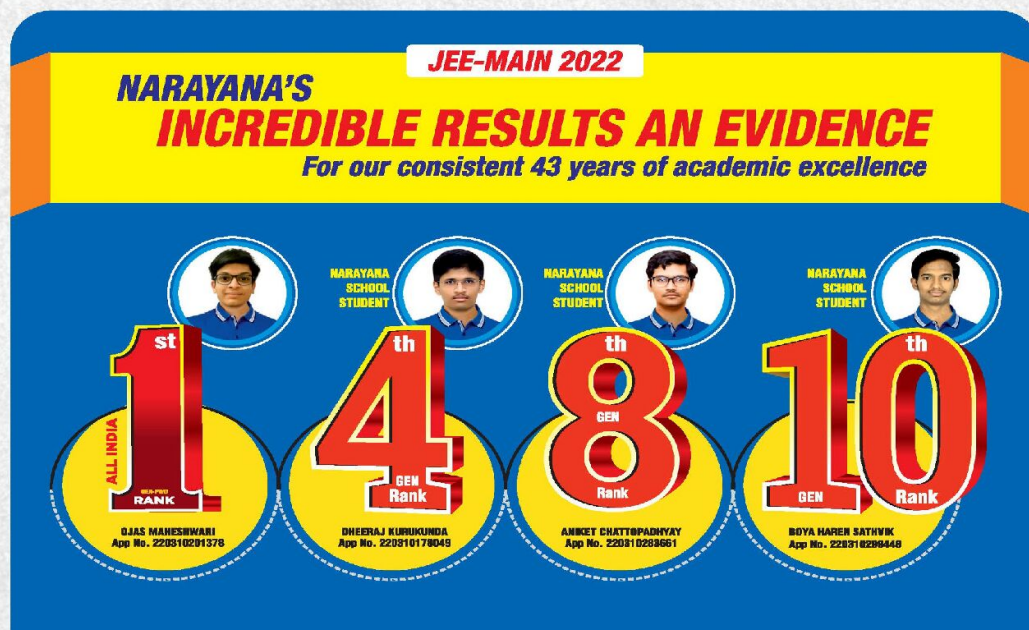


NARAYANA
IIT-JEE ACADEMY - INDIA

NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
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ORGANIC CHEMISTRY PART-I (JR) ADVANCED MODEL QUESTIONS



PHYSICS

SHM, WAVES & OPTICS

JEE ADVANCED IMPORTANT QUESTIONS

44
YEARS
OF EXCELLENCE



NARAYANA
IIT-JEE ACADEMY-INDIA

1. For a plane electromagnetic wave, the magnetic field at a point x and time t is E_0
 The instantaneous electric field $\vec{E}$ corresponding to $\vec{B}$ is:
 (speed of light $c = 3 \times 10^8 \text{ ms}^{-1}$)
 (A) $\vec{E}(x, t) = [-36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{j}] \frac{V}{m}$
 (B) $\vec{E}(x, t) = [36 \sin(1 \times 10^3 x + 0.5 \times 10^{11} t) \hat{j}] \frac{V}{m}$
 (C) $\vec{E}(x, t) = [36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{k}] \frac{V}{m}$
 (D) $\vec{E}(x, t) = [36 \sin(1 \times 10^3 x + 1.5 \times 10^{11} t) \hat{i}] \frac{V}{m}$
2. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0(\hat{x} + \hat{y}) \sin(kz - \omega t)$
 Its magnetic field will be given by:
 (A) $\frac{E_0}{c}(-\hat{x} + \hat{y}) \sin(kz - \omega t)$ (B) $\frac{E_0}{c}(\hat{x} + \hat{y}) \sin(kz - \omega t)$
 (C) $\frac{E_0}{c}(\hat{x} - \hat{y}) \sin(kz - \omega t)$ (D) $\frac{E_0}{c}(\hat{x} - \hat{y}) \cos(kz - \omega t)$
3. The electric field of a plane electromagnetic wave propagating along the x direction in vacuum is $\vec{E} = E_0 \hat{j} \cos(\omega t - kx)$. The magnetic field $\vec{B}$, at the moment $t = 0$ is
 (A) $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \epsilon_0}} \cos(kx) \hat{k}$ (B) $\vec{B} = E_0 \sqrt{\mu_0 \epsilon_0} \cos(kx) \hat{j}$
 (C) $\vec{B} = E_0 \sqrt{\mu_0 \epsilon_0} \cos(kx) \hat{k}$ (D) $\vec{B} = \frac{E_0}{\sqrt{\mu_0 \epsilon_0}} \cos(kx) \hat{j}$
4. In plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}$ and $2\hat{i} - 2\hat{j}$, respectively. What is the unit vector along direction of propagation of the wave
 (A) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ (B) $\frac{1}{\sqrt{2}}(\hat{j} + \hat{k})$ (C) $\frac{1}{\sqrt{5}}(\hat{i} + 2\hat{j})$ (D) $\frac{1}{\sqrt{5}}(2\hat{i} + \hat{j})$
5. A plane electromagnetic wave is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form of the possible magnetic field of the wave at some point would be (here B_0 is an appropriate positive constant).
 (A) $B_0 \frac{\hat{i} - \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$ (B) $B_0 \frac{\hat{j} - \hat{i}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$
 (C) $B_0 \hat{k} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$ (D) $B_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos\left(\omega t - k \frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$

6. If the magnetic field in a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{j} \text{ T}$, then what will be expression for electric field?
- (A) $\vec{E} = (60 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m})$
 (B) $\vec{E} = (9 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m})$
 (C) $\vec{E} = (3 \times 10^{-8} (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m})$
 (D) $\vec{E} = (3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m})$
7. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos(kz + \omega t)$. At $t = 0$, a positively charged particle is at the point $(x, y, z) = \left(0, 0, \frac{\pi}{k}\right)$. If its instantaneous velocity at $(t = 0)$ is $v_0 \hat{k}$, the force acting on it due to the wave is: [E_0 is a positive constant]
- (A) parallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (B) zero
 (C) antiparallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (D) parallel to $\hat{k}$
8. A plane electromagnetic wave having a frequency $\nu = 23.9 \text{ GHz}$ propagates along the positive z – direction in free space. The peak value of the Electric Field is 60 V/m . Which among the following is the acceptable magnetic field component in the electromagnetic wave?
- (A) $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z + 1.5 \times 10^{11} t) \hat{i}$
 (B) $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$
 (C) $\vec{B} = 60 \sin(0.5 \times 10^3 x - 1.5 \times 10^{11} t) \hat{k}$
 (D) $\vec{B} = 2 \times 10^{-7} \sin(1.5 \times 10^2 x - 0.5 \times 10^{11} t) \hat{j}$
9. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \hat{i} \cos(kz) \cos(\omega t)$. The corresponding magnetic field is then given by: [C is velocity of light]
- (A) $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \sin(\omega t)$ (B) $\vec{B} = \frac{E_0}{C} \hat{j} \sin(kz) \cos(\omega t)$
 (C) $\vec{B} = \frac{E_0}{C} \hat{j} \cos(kz) \sin(\omega t)$ (D) $\vec{B} = \frac{E_0}{C} \hat{k} \sin(kz) \cos(\omega t)$
10. 50 W/m^2 intensity of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on 1 m^2 surface area will be close to ($c = 3 \times 10^8 \text{ m/s}$):

- (A) $15 \times 10^{-8} \text{ N}$ (B) $20 \times 10^{-8} \text{ N}$ (C) $10 \times 10^{-8} \text{ N}$ (D) $35 \times 10^{-8} \text{ N}$

11. The magnetic field of an electromagnetic wave is given by:

$$\vec{B} = 1.6 \times 10^{-6} \cos(2 \times 10^7 z + 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{Wb}}{\text{m}^2}$$
 The associated electric field will be:

- (A) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{V}}{\text{m}}$
 (B) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z - 6 \times 10^{15} t) (-2\hat{i} + \hat{j}) \frac{\text{V}}{\text{m}}$
 (C) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (-\hat{i} + 2\hat{j}) \frac{\text{V}}{\text{m}}$
 (D) $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (\hat{i} - 2\hat{j}) \frac{\text{V}}{\text{m}}$

12. A 27 mW laser beam has a cross – sectional area of 10 mm^2 . The magnitude of the maximum electric field in this electromagnetic wave is given by:

[Given permittivity of space $\epsilon_0 = 9 \times 10^{-12} \text{ SI units}$, speed of light $c = 3 \times 10^8 \text{ m/s}$]

- (A) 2kV/m (B) 0.7kV/m (C) 1kV/m (D) 1.4kV/m

13. The electric field of a plane polarized electromagnetic wave in free space at time $t = 0$ is given by an expression $\vec{E}(x, y) = 10j \cos[(6x + 8z)]$ The magnetic field $\vec{B}(x, z, t)$ is given by: (c is the velocity of light)

- (A) $\frac{1}{c} (6\hat{k} + 8\hat{i}) \cos[(6x - 8z + 10ct)]$ (B) $\frac{1}{c} (6\hat{k} + 8\hat{i}) \cos[(6x + 8z - 10ct)]$
 (C) $\frac{1}{c} (6\hat{k} + 8\hat{i}) \cos[(6x + 8z - 10ct)]$ (D) $\frac{-1}{c} (6\hat{k} - 8\hat{i}) \cos[(6x + 8z + 10ct)]$

14. A monochromatic beam of light has a frequency $\nu = \frac{3}{2\pi} \times 10^{12} \text{ Hz}$ and is propagating along the direction $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$. It is polarized along the $\hat{k}$ direction. The acceptable form for the magnetic field is:

- (A) $k \frac{E_0}{C} \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}} \right) \cos \left[10^4 \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}} \right) \cdot \vec{r} - (3 \times 10^{12})t \right]$
 (B) $\frac{E_0}{C} \left(\frac{\hat{i} - \hat{j}}{\sqrt{2}} \right) \cos \left[10^4 \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \cdot \vec{r} - (3 \times 10^{12})t \right]$
 (C) $\frac{E_0}{C} \hat{k} \cos \left[10^4 \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \cdot \vec{r} + (3 \times 10^{12})t \right]$
 (D) $\frac{E_0}{C} \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}} \cos \left[10^4 \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \cdot \vec{r} + (3 \times 10^{12})t \right]$

15. The electric field component of a monochromatic radiation is given by $\vec{E} = 2E_0 \hat{i} \cos kz \cos \omega t$
Its magnetic field $\vec{B}$ is then given by:
- (A) $\frac{2E_0}{c} \hat{j} \sin kz \cos \omega t$ (B) $-\frac{2E_0}{c} \hat{j} \sin kz \sin \omega t$
(C) $\frac{2E_0}{c} \hat{j} \sin kz \sin \omega t$ (D) $\frac{2E_0}{c} \hat{j} \cos kz \cos \omega t$
16. A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 W of power. The amplitude of the electric field associated with the electromagnetic radiation at a distance of 5m from the lamp will be nearly.
- (A) 1.34 V / m (B) 2.68 V / m (C) 4.02 V / m (D) 5.36 V / m
17. Arrange the following electromagnetic radiations per quantum in the order of increasing energy:
A : Blue light B : Yellow light C : X – ray D : Radio wave
(A) C, A, B, D (B) B, A, D, C (C) D, B, A, C (D) A, B, D, C
18. Microwave oven acts on the principle of:
- (A) Giving rotational energy to water molecules
(B) Giving translational energy to water molecules
(C) Giving vibrational energy to water molecules
(D) transferring electrons from lower to higher energy levels in water molecules.
19. If microwaves, X rays, infrared, gamma rays, ultra – violet, radio waves and visible parts of the electromagnetic spectrum are denoted by M, X, I, G U, R and V then which of the following is the arrangement in ascending order of wavelength?
- (A) R, M, I, V, U, X and G (B) M, R, V, X, U, G and I
(C) G, X, U, V, I, M and R (D) I, M, R, U, V, X and G
20. Photons of an electromagnetic radiation have an energy 11 keV each. To which region of electromagnetic spectrum does it belong?
- (A) X – rays region (B) Ultra violet region
(C) Infrared region (D) Visible region

21. The correct match between the entries in column I and column II are:

	I Radiation		II Wavelength
(a)	Microwave	(i)	100 m
(b)	Gamma rays	(ii)	10^{-15} m
(c)	A.M. radio waves	(iii)	10^{-10} m
(d)	X-rays	(iv)	10^{-3} m

- (A) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii) (B) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)
 (C) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv) (D) (a) – (iv), (b) – (ii), (c) – (i), (d) – (iii)

22. Given below in the left column are different modes of communication using the kinds of waves given in the right column.

a.	Optical Fibre	p.	Ultrasound
b.	Radar	q.	Infrared Light
c.	Sonar	r.	Microwaves
d.	Mobile Phones	s.	Radio Waves

From the options given below, find the most appropriate match between entries in the left and the right column.

- (A) a – q, b – s, c – r, d – p (B) a – s, b – q, c – r, d – p
 (C) a – q, b – s, c – p, d – r (D) a – r, b – p, c – s, d – q
23. Match List – I (Electromagnetic wave type) with List – II (Its association / application) and select the correct option from the choices given below the lists:

	List – I		List – II
1.	Infrared waves	(i)	To treat muscular strain
2.	Radio waves	(ii)	For broadcasting
3.	X-rays	(iii)	To detect fracture of bones
4.	Ultraviolet rays	(iv)	Absorbed by the ozone layer of the atmosphere

24. Match the List – I (Phenomenon associate with electromagnetic radiation) with List – II (Part of electromagnetic spectrum) and select the most appropriate code from the choices given below this lists:

	List – I		List – II
I	Doublet of sodium	(a)	Visible radiation
II	Wavelength corresponding to temperature associated with the isotropic radiation filling all space (cosmic background radiation)	(b)	Microwave
III	Wavelength emitted by atomic hydrogen in interstellar space	(c)	Short radio wave
IV	Wavelength of radiation arising from two close energy levels in hydrogen (balmer series)	(d)	X – rays

- (A) (I) – (a), (II) – (b), (III) – (b), (IV) – (a)
 (B) (I) – (a), (II) – (b), (III) – (c), (IV) – (c)
 (C) (I) – (d), (II) – (c), (III) – (a), (IV) – (a)
 (D) (I) – (b), (II) – (a), (III) – (a), (IV) – (a)
25. Match List I (Wavelength range of electromagnetic spectrum) with List II (Method of production of these waves) and select the correct option from the options given below the lists.

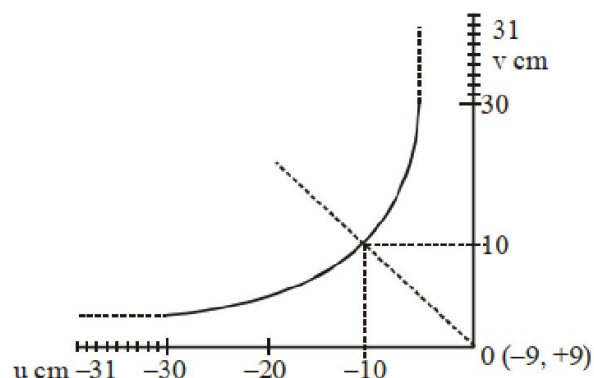
	List – I		List – II
(1)	700 nm to 1 mm	(i)	Vibration of atom and molecules.
(2)	1 nm to 400 nm	(ii)	Electrons in atoms moving from one energy level to lower level.
(3)	$< 10^{-3}$ nm	(iii)	Radioactive decay of the nucleus
(4)	1 mm to 0.1 m	(iv)	Magnetron valve.

- (A) (1) – (iv), (2) – (iii), (3) – (ii), (4) – (i)
 (B) (1) – (iii), (2), (iv), (3) – (i), (4) – (ii)

(C) (1) – (ii), (2) – (iii), (3) – (iv), (4) – (i)

(D) (1) – (i), (2) – (ii), (3) – (iii), (4) – (iv)

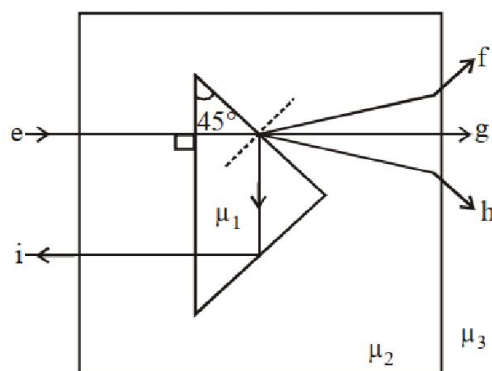
26. The graph shows relationship between object distance and image distance for a equiconvex lens. Then, focal length of the lens is



- (A) $0.50 \pm 0.05\text{cm}$ (B) $0.50 \pm 0.10\text{cm}$ (C) $5.00 \pm 0.05\text{cm}$ (D) $5.00 \pm 0.10\text{cm}$
27. A thin prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of the prism P_2 is
- (A) 5.33° (B) 4° (C) 3° (D) 2.6°
28. A right-angled prism of refractive index μ_1 is placed in a rectangular block of refractive index μ_2 , which is surrounded by a medium of refractive index μ_3 , as shown in the figure. A ray of light 'e' enters of rectangular block at normal incidence. Depending upon the relationships between μ_1, μ_2 and μ_3 , it takes one of the four possible paths 'ef', 'eg', 'eh' or 'ei'.

Match the paths in List I with conditions of refractive indices in List II and select the correct answer using the codes given below the lists:

	List – I		List – II
P.	$e \rightarrow f$	1.	$\mu_1 > \sqrt{2}\mu_2$
Q.	$e \rightarrow g$	2.	$\mu_2 > \mu_1$ and $\mu_2 > \mu_3$
R.	$e \rightarrow h$	3.	$\mu_1 = \mu_2$
S.	$e \rightarrow i$	4.	$\mu_2 < \mu_1 < \sqrt{2}\mu_2$ and $\mu_2 > \mu_3$

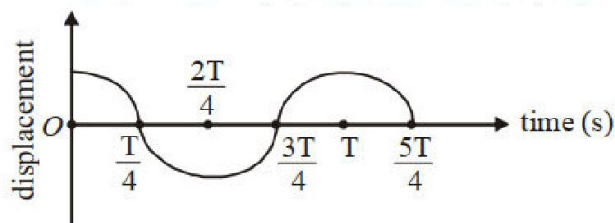


Code:

	P	Q	R	S
(a)	2	3	1	4
(b)	1	2	4	3
(c)	4	1	2	3
(d)	2	3	4	1

29. The magnifying power of a telescope with tube length 60 cm is 5. What is the focal length of its eye piece?
- (A) 20 cm (B) 40 cm (C) 30 cm (D) 10 cm
30. If we need a magnification of 375 from a compound microscope of tube length 150 mm and an objective of focal length 5 cm, the focal length of the eyepiece, should be close to:
- (A) 22 mm (B) 12 mm (C) 2 mm (D) 33 mm
31. An observer looks at a distant tree of height 10m with a telescope of magnifying power of 20. To the observer the tree appears:
- (A) 20 times taller (B) 20 times nearer (C) 10 times taller (D) 10 times nearer
32. To determine refractive index of glass slab using a travelling microscope, minimum number of readings required are:
- (A) Two (B) Four (C) Three (D) Five
33. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. If a 50 m tall tower at a distance of 1 km is observed through this telescope in normal setting, the angle formed by the image of the tower is θ , then θ is closest to:
- (A) 30° (B) 15° (C) 84° (D) 1°
34. In a compound microscope, the focal length of objective lens is 1.2 cm and focal length of objective lens is 1.2 cm and focal length of eye piece is 3.0 cm. when object is kept at 1.25 cm in front of objective, final image is formed at infinity. Magnifying power of the compound microscope should be:
- (A) 200 (B) 100 (C) 400 (D) 150

35. The focal lengths of the objective and the eye piece of a compound microscope are 2.0 cm and 3.0 cm, respectively. The distance between the objective and the eye piece is 15.0 cm. The final image formed by the eye piece is at infinity. The two lenses are thin, The distance in cm of the object and the image produced by the objective, measured from the objective lens, are respectively
 (A) 2.4 and 12.0 (B) 2.4 and 15.0 (C) 2.0 and 12.0 (D) 2.0 and 3.0
36. A compound microscope consists of an objective lens of focal length 1 cm and an eyepiece of focal length 5 cm with a separation of 10 cm.
 The distance between an object and the objective lens, at which the strain on the eye is minimum is $\frac{n}{40}$ cm. The value of n is _____.
37. In a compound microscope, the magnified virtual image is formed at a distance of 25 cm from the eye – piece. The focal length of its objective lens is 1 cm. If the magnification is 100 and the tube length of the microscope is 20 cm, then the focal length of the eyepiece lens (in cm) is _____.
38. A planet is observed by an astronomical refracting telescope having an objective of focal length 16 m and an eyepiece of focal length 2 cm.
 (A) The distance between the objective and the eyepiece is 16.02 m
 (B) The angular magnification of the planet is –800
 (C) The image of the planet is inverted
 (D) The objective is larger than the eyepiece
39. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and the eyepiece is 36 cm and the final image is formed at infinity. The focal length f_o of the objective and f_e of the eyepiece are
 (A) $f_o = 45\text{cm}$ and $f_e = -9\text{cm}$ (B) $f_o = 50\text{cm}$ and $f_e = 10\text{cm}$
 (C) $f_o = 7.2\text{cm}$ and $f_e = 5\text{cm}$ (D) $f_o = 30\text{cm}$ and $f_e = 6\text{cm}$
40. The displacement time graph of a particle executing S.H.M. is given in figure: (sketch is schematic and not to scale)



Which of the following statements is/are true for this motion?

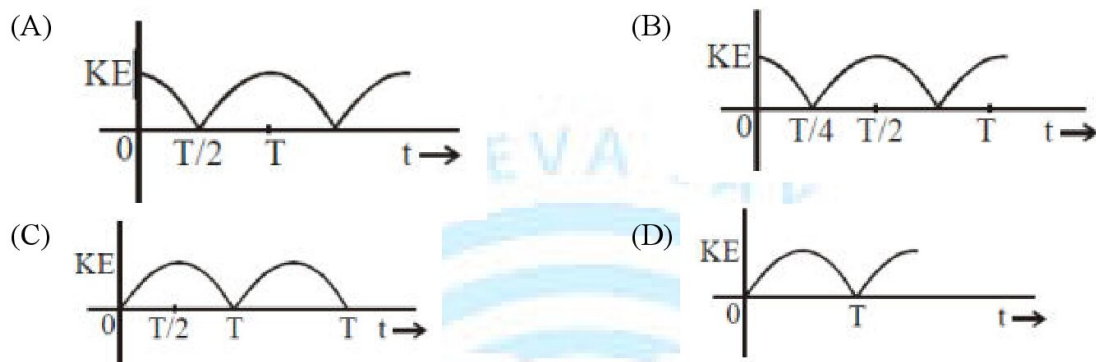
- (1) The force is zero at $t = \frac{3T}{4}$
 (2) The acceleration is maximum at $t = T$

(3) The speed is maximum at $t = \frac{T}{4}$

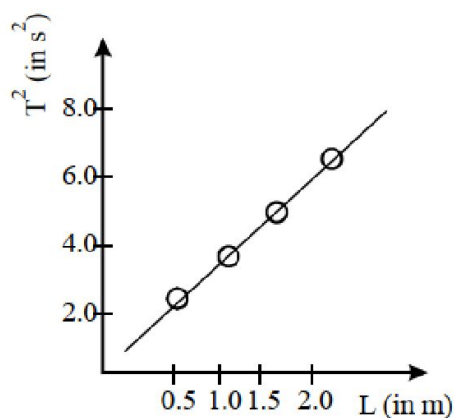
(4) The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$

(A) (1), (2) and (4) (B) (2), (3) and (4) (C) (1), (2) and (3) (D) (1) and (4)

41. A particle is executing simple harmonic motion with a time period T . At time $t = 0$, it is at its position of equilibrium. The kinetic energy – time graph of the particle will look like:



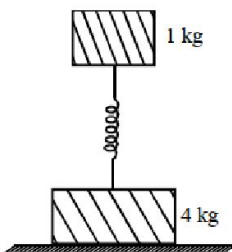
42. An object of mass 0.2 kg executes simple harmonic oscillation along the x – axis with a frequency of $(25/\pi) \text{ Hz}$. At the position $x = 0.04$, the object has Kinetic energy of 0.5 J and potential energy 0.4 J . The amplitude of oscillations ism.
43. In an experiment for determining the gravitational acceleration g of a place with the help of a simple pendulum, the measured time period square is plotted against the string length of the pendulum in the figure.



What is the value of g at the place?

(A) 9.81 m/s^2 (B) 9.87 m/s^2 (C) 9.91 m/s^2 (D) 10.0 m/s^2

44. Two bodies of masses 1 kg and 4 kg are connected to a vertical spring, as shown in the figure. The smaller mass executes simple harmonic motion of angular frequency 25 rad/s, and amplitude 1.6 cm while the bigger mass remains stationary on the ground. The maximum force exerted by the system on the floor is (take $g = 10 \text{ ms}^{-2}$)



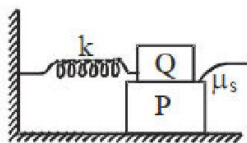
- (A) 20N (B) 10N (C) 60N (D) 40N
45. An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M . The piston and the cylinder have equal cross sectional area A . When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0 . The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency

(A) $\frac{1}{2\pi} \frac{A\gamma P_0}{V_0 M}$ (B) $\frac{1}{2\pi} \frac{V_0 M P_0}{A^2 \gamma}$ (C) $\frac{1}{2\pi} \sqrt{\frac{A^2 \gamma P_0}{M V_0}}$ (D) $\frac{1}{2\pi} \sqrt{\frac{M V_0}{A \gamma P_0}}$

46. Two simple pendulums of length 1 m and 4 m respectively are both given small displacement in the same direction at the same instant. They will be again in phase after the shorter pendulum has completed number of oscillations equal to:
- (A) 2 (B) 7 (C) 5 (D) 3
47. A simple pendulum has time period T_1 . The point of suspension is now moved upward according to the relation $y = Kt^2$, ($K = 1 \text{ m/s}^2$) where y is the vertical displacement.

The time period now becomes T_2 . The ratio of $\frac{T_1^2}{T_2^2}$ is ($g = 10 \text{ m/s}^2$)

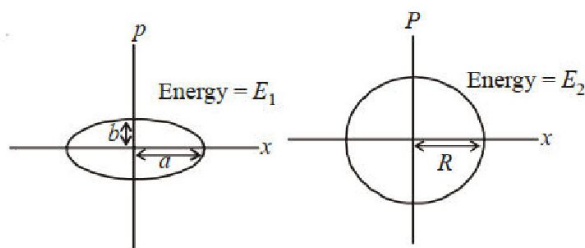
- (A) 5/6 (B) 6/5 (C) 1 (D) 4/5
48. A block P of mass m is placed on a horizontal frictionless plane. A second block of same mass m is placed on it and is connected to a spring of spring constant k , the two blocks are pulled by distance A . Block Q oscillates without slipping. What is the maximum value of frictional force between the two blocks.



- (A) $kA/2$ (B) kA (C) $\mu_s mg$ (D) zero

49. A particle of mass m is executing oscillations about the origin on the x axis. Its potential energy is $V(x) = k|x|^3$ where k is a positive constant. If the amplitude of oscillation is a , then its time period T is
- (A) Proportional to $1/\sqrt{a}$ (B) Independent of a
 (C) Proportional to $\sqrt{a}$ (D) Proportional to $a^{3/2}$
50. A highly rigid cubical block A of small mass M and side L is fixed rigidly on to another cubical block B of the same dimensions and of low modulus of rigidity η such that the lower face of A completely covers the upper face of B. The lower face of B is rigidly held on a horizontal surface. A small force F is applied perpendicular to one of the sides faces of A. After the force is withdrawn, block A executes small oscillations the time period of which is given by
- (A) $2\pi\sqrt{M\eta L}$ (B) $2\pi\sqrt{\frac{M\eta}{L}}$ (C) $2\pi\sqrt{\frac{ML}{\eta}}$ (D) $2\pi\sqrt{\frac{M}{\eta L}}$
51. A uniform cylinder of length L and mass M having cross sectional area A is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half – submerged in a liquid of density ρ at equilibrium position. When the cylinder is given a small downward push and released it starts oscillating vertically with small amplitude. If the force constant of the spring is k , the frequency of oscillation of the cylinder is
- (A) $\frac{1}{2\pi} \left(\frac{k - A\rho g}{M} \right)^{1/2}$ (B) $\frac{1}{2\pi} \left(\frac{k + A\rho g}{M} \right)^{1/2}$
 (C) $\frac{1}{2\pi} \left(\frac{k + \rho g L}{M} \right)^{1/2}$ (D) $\frac{1}{2\pi} \left(\frac{k + A\rho g}{A\rho g} \right)^{1/2}$
52. A particle of mass m is attached to one end of a massless spring of force constant k , lying on a frictionless horizontal plane. The other end of the spring is fixed. The particle starts moving horizontally from its equilibrium position at time $t = 0$ with an initial velocity u_0 . When the speed of the particle is $0.5u_0$, it collides elastically with a rigid wall. After this collision
- (A) The speed of the particle when it returns to its equilibrium position is u_0
 (B) The time at which the particle passes through the equilibrium position for the first time is $t = \pi\sqrt{\frac{m}{k}}$
 (C) The time at which the maximum compression of the spring occurs is $t = \frac{4\pi}{3}\sqrt{\frac{m}{k}}$
 (D) The time at which the particle passes through the equilibrium position for the second time is $t = \frac{5\pi}{3}\sqrt{\frac{m}{k}}$

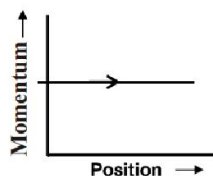
53. A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original amplitude is close to:
 (A) 50s (B) 100s (C) 20s (D) 10s
54. An oscillator of mass M is at rest in its equilibrium position in a potential $V = \frac{1}{2}k(x - X)^2$. A particle of mass m comes from right with speed u and collides completely inelastically with M and sticks to it. This process repeats every time the oscillator crosses its equilibrium position. The amplitude of oscillations after 13 collisions is:
 ($M = 10, m = 5, u = 1, k = 1$)
 (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{2}{3}$ (D) $\sqrt{\frac{3}{5}}$
55. The angular frequency of the damped oscillator is given by, $\omega = \sqrt{\left(\frac{k}{m} - \frac{r^2}{4m^2}\right)}$ where k is the spring constant, m is the mass of the oscillator and r is the damping constant. If the ratio $\frac{r^2}{mk}$ is 8%, the change in time period compared to the undamped oscillator is approximately as follows:
 (A) increase by 1% (B) increase by 8% (C) decrease by 1% (D) decreases by 8%
56. The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5s. In another 10s it will decrease to α times its original magnitude, where α equals
 (A) 0.7 (B) 0.81 (C) 0.729 (D) 0.6
57. A 0.1 kg mass is suspended from a wire of negligible mass. The length of the wire is 1m and its cross-sectional area $4.9 \times 10^{-7} \text{ m}^2$. If the mass is pulled a little in the vertically downward direction and released, it performs simple harmonic motion of angular frequency 140 rad s^{-1} . If the Young's modulus of the material of the wire is $n \times 10^9 \text{ Nm}^{-2}$, the value of n is
58. Two independent harmonic oscillators of equal mass are oscillating about the origin with angular frequencies ω_1 and ω_2 and have total energies E_1 and E_2 , respectively. The variations of their momenta p with positions x are shown in the figures. If $\frac{a}{b} = n^2$ and $\frac{a}{R} = n$, then the correct equation(s) is(are)



- $E_1\omega_1 = E_2\omega_2$ (A) (B) $\frac{\omega_2}{\omega_1} = n^2$ (C) $\omega_1\omega_2 = n^2$ (D) $\frac{E_1}{\omega_1} = \frac{E_2}{\omega_2}$

Passage – 1 (59-61)

Phase space diagrams are useful tools in analysing all kinds of dynamical problems. They are especially useful in studying the changes in motion as initial position and momentum are changed. Here we consider some simple dynamical systems in one dimension.

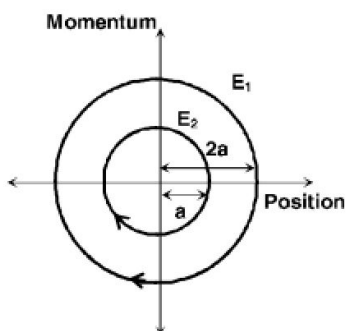


For such systems, phase space is a plane in which position is plotted along horizontal axis and momentum is plotted along vertical axis. The phase space diagram is $x(t)$ vs. $p(t)$ curve in this plane. The arrow on the curve indicates the time flow. For example, the phase space diagram for a particle moving with constant velocity is a straight line as shown in the figure. We use the sign convention in which position or momentum upwards (or to right) is positive and downwards (or to left) is negative.

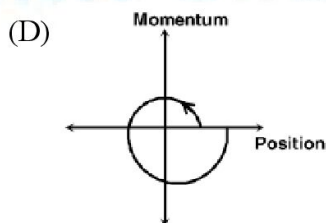
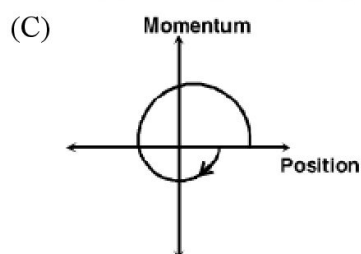
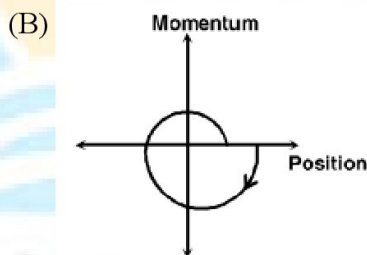
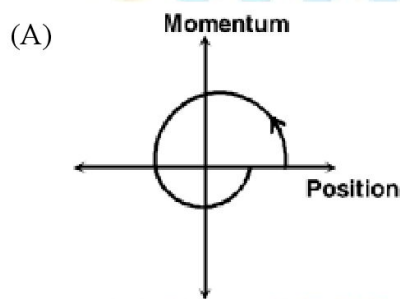
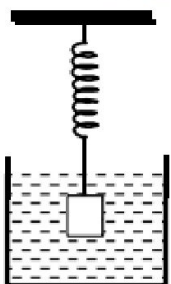
59. The phase space diagram for a ball thrown vertically up from ground is

- (A) (B)
- (C) (D)

60. The phase space diagram for simple harmonic motion is a circle centered at the origin. In the figure, the two circles represent the same oscillator but for different initial conditions, and E_1 and E_2 are the total mechanical energies respectively. Then

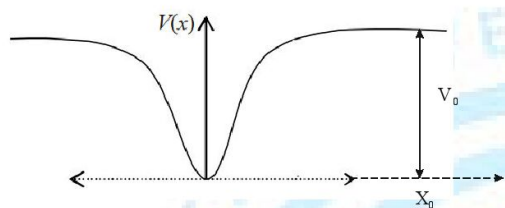


- (A) $E_1 = \sqrt{2}E_2$ (B) $E_1 = 2E_2$ (C) $E_1 = 4E_2$ (D) $E_1 = 16E_2$
61. Consider the spring – mass system, with the mass submerged in water, as shown in the figure. The phase space diagram for one cycle of this system is



Passage - 2

When a particle of mass m moves on the x – axis in a potential of the form $V(x) = kx^2$ it performs simple harmonic motion. The corresponding time period is proportional to $\sqrt{\frac{m}{k}}$, as can be seen easily using dimensional analysis. However, the motion of a particle can be periodic even when its potential energy increase on both sides of $x = 0$ in a way different from kx^2 and its total energy is such that the particle does not escape to infinity. Consider a particle of mass m moving on the x – axis. Its potential energy is $V(x) = \alpha x^4$ ($\alpha > 0$) for $|x|$ near the origin and becomes a constant equal to V_0 for $|x| \geq X_0$ (see figure).



62. If the total energy of the particle is E , it will perform periodic motion only if
 (A) $E < 0$ (B) $E > 0$ (C) $V_0 > E > 0$ (D) $E > V_0$
63. For periodic motion of small amplitude A , the time period T of this particle is proportional to
 (A) $A\sqrt{\frac{m}{\alpha}}$ (B) $\frac{1}{A}\sqrt{\frac{m}{\alpha}}$ (C) $A\sqrt{\frac{\alpha}{m}}$ (D) $\frac{1}{A}\sqrt{\frac{\alpha}{m}}$
64. The acceleration of this particle for $|x| > X_0$ is
 (A) Proportional to V_0 (B) Proportional to $\frac{V_0}{mX_0}$
 (C) Proportional to $\sqrt{\frac{V_0}{mX_0}}$ (D) Zero
65. Unpolarized light of intensity I is incident on a system of two polarizers, A followed by B. The intensity of emergent light is $I/2$. If a third polarizer C is placed between A and B, the intensity of emergent light is reduced to $I/3$. The angle between the polarizers A and C is θ . Then
 (A) $\cos \theta = \left(\frac{2}{3}\right)^{1/4}$ (B) $\cos \theta = \left(\frac{1}{3}\right)^{1/4}$ (C) $\cos \theta = \left(\frac{1}{3}\right)^{1/2}$ (D) $\cos \theta = \left(\frac{2}{3}\right)^{1/2}$
66. An observer is moving with half the speed of light towards a stationary microwave source emitting waves at frequency 10 GHz. What is the frequency of the microwave measured by the observer?
 (speed of light = $3 \times 10^8 \text{ ms}^{-1}$)
 (A) 17.3GHz (B) 15.3GHz (C) 10.1GHz (D) 12.1GHz

67. Two beams, A and B, of plane polarized light with mutually perpendicular planes of polarization are seen through a polaroid. From the position when the beam A has maximum intensity (and beam B has zero intensity), a rotation of polaroid through 30° makes the two beams appear equally bright. If the initial intensities of the two beams are I_A and I_B respectively, then $\frac{I_A}{I_B}$ equals:

(A) 3 (B) $\frac{3}{2}$ (C) 1 (D) $\frac{1}{3}$

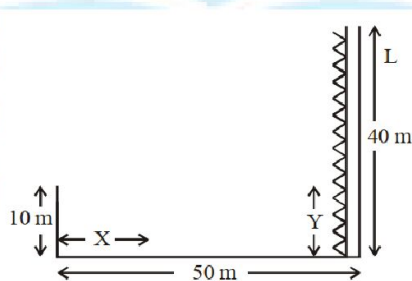
68. A ray of light is incident from a denser to a rarer medium. The critical angle for total internal reflection is θ_{ic} and Brewster's angle of incidence is θ_{ib} , such that $\sin \theta_{ic} / \sin \theta_{ib} = \eta = 1.28$. The relative refractive index of the two media is:

(A) 0.2 (B) 0.4 (C) 0.8 (D) 0.9

69. In an experiment of single slit diffraction pattern, first minimum for red light coincides with first maximum of some other wavelength. If wavelength of red light is $6600 \text{ \AA}$, then wavelength of other light will be:

(A) $3300 \text{ \AA}$ (B) $4400 \text{ \AA}$ (C) $5500 \text{ \AA}$ (D) $6600 \text{ \AA}$

70. A person lives in a high-rise building on the bank of a river 50m wide. Across the river is a well lit tower of height 40 m. When the person, who is at a height of 10m, looks through a polarizer at an appropriate angle at light of the tower reflecting from the river surface, he notes that intensity of light coming from distance X from his building is the least and this corresponds to the light coming from light bulbs at height 'Y' on the tower. The values of X and Y are respectively close to (refractive index of water $\approx \frac{4}{3}$)



- (A) 25m, 10 m (B) 13m, 27m (C) 22m, 13m (D) 17m, 20m
71. Yellow light is used in a single slit diffraction experiment with slit width of 0.6mm. If yellow light is replaced by X-rays, then the observed pattern will reveal,
- (A) That the central maximum is narrower
 (B) More number of fringes
 (C) Less number of fringes
 (D) No diffraction pattern

72. Consider Fraunhofer diffraction pattern obtained with a single slit illuminated at normal incidence. At the angular position of the first diffraction minimum the phase difference (in radians) between the wavelets from the opposite edges of the slit is
 (A) $\pi/4$ (B) $\pi/2$ (C) 2π (D) π
73. A beam of light of wave length 600 nm from a distant source falls on a single slit 1 mm wide and a resulting diffraction pattern is observed on a screen 2m away. The distance between the first dark fringes on either side of central bright fringe is
 (A) 1.2cm (B) 1.2mm (C) 2.4cm (D) 2.4mm
74. A parallel monochromatic beam of light is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of the incident beam. At the first minimum of the diffraction pattern, the phase difference between the rays coming from the two edges of the slit is
 (A) 0 (B) $\pi/2$ (C) π (D) 2π
75. Orange light of wavelength 6000×10^{-10} m illuminates a single slit of width 0.6×10^{-4} m. The maximum possible number of diffraction minima produced on both sides of the central maximum is _____
76. A simple telescope used to view distance objects has eyepiece and objective lens of focal lengths f_e and f_o , respectively Then

	Column – I		Column – II
(a)	Amount of light received by lens	(p)	Radius of aperture
(b)	Angular magnification	(q)	Dispersion of lens
(c)	Length of telescope tube	(r)	Focal length of objective lens and eyepiece lens
(d)	Sharpness of image	(s)	Spherical aberration

77. A sonometer wire of length 114 cm is fixed at both the ends. Where should the two bridges be placed so as to divide the wire into three segments whose fundamental frequencies are in the ratio 1 : 3 : 4?
 (A) At 36 cm and 84 cm from one end
 (B) At 24 cm and 72 cm from one end
 (C) At 48 cm and 96 cm from one end
 (D) At 72 cm and 96 cm from one end

78. An open pipe is in resonance in 2nd harmonic with frequency f_1 . Now one end of the tube is closed and frequency is increased to f_2 such that the resonance again occurs in n th harmonic. Choose the correct option.
- (A) $n = 3, f_2 = \frac{3}{4}f_1$ (B) $n = 3, f_2 = \frac{5}{4}f_1$ (C) $n = 5, f_2 = \frac{3}{4}f_1$ (D) $n = 5, f_2 = \frac{5}{4}f_1$
79. A 20cm long string, having a mass of 1.0g, is fixed at both the ends. The tension in the string is 0.5. The string is set into vibrations using an external vibrator of frequency 100Hz. Find the separation (in cm) between the successive nodes on the string.
80. A wire of density $9 \times 10^{-3} \text{ kg cm}^{-3}$ is stretched between two clamps 1m apart. The resulting strain in the wire is 4.9×10^{-4} . The lowest frequency of the transverse vibrations in the wire is (Young's modulus of wire $Y = 9 \times 10^{10} \text{ Nm}^{-2}$), (to the nearest integer), _____.
81. One end of a taut string of length 3 m along the x – axis is fixed at $x = 0$. The speed of the waves in the string is 100 ms^{-1} . The other end of the string is vibrating in the y – direction so that stationary waves are set up in the string. The possible waveform (s) of these stationary waves is (are)
- (A) $y(t) = A \sin \frac{\pi x}{6} \cos \frac{50\pi t}{3}$ (B) $y(t) = A \sin \frac{\pi x}{3} \cos \frac{100\pi t}{3}$
- (C) $y(t) = A \sin \frac{5\pi x}{6} \cos \frac{250\pi t}{3}$ (D) $y(t) = A \sin \frac{5\pi x}{6} \cos 250\pi t$
82. A musical instrument is made using four different metal strings 1, 2, 3 and 4 with mass per unit length $\mu, 2\mu, 3\mu$ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$. It is found that in string – 1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0 . [strings are fixed at both ends]

List – I gives the above four strings while list – II lists the magnitude of some quantity.

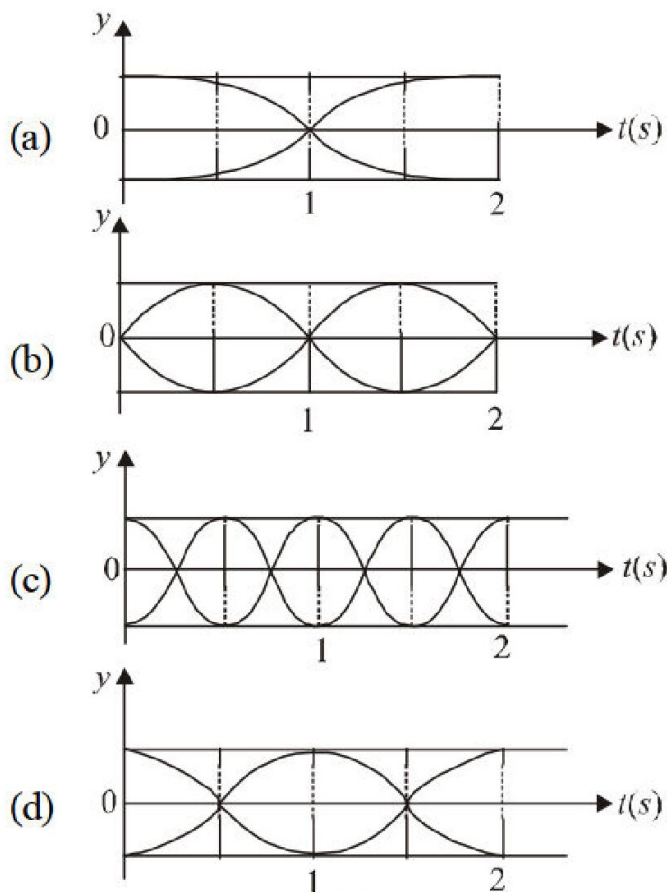
	List – I		List – II
(I)	String – 1 (μ)	(P)	1
(II)	String – 2 (2μ)	(Q)	$1/2$
(III)	String – 3 (3μ)	(R)	$1/\sqrt{2}$
(IV)	String – 4 (4μ)	(S)	$1/\sqrt{3}$
		(T)	$3/16$
		(U)	$1/16$

The length of the strings 1, 2, 3 and 4 are kept fixed at $L_0, \frac{3L_0}{2}, \frac{5L_0}{4}$, and $\frac{7L_0}{4}$, respectively. Strings

1, 2, 3, and 4 are vibrated at their 1st, 3rd, 5th, and 14th harmonics, respectively such that all the strings have same frequency. The correct match for the tension in the four strings in the units of T_0 will be

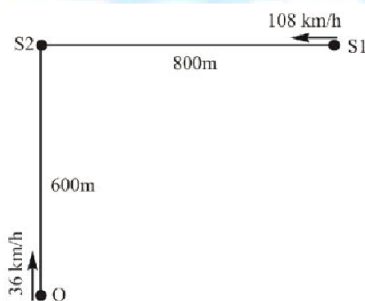
- (A) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ U
 (B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ U
 (C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ T
 (D) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ U

83. A tuning fork of frequency 480Hz resonates with a tube closed at one end of length 16 cm and diameter 5cm in fundamental mode. Calculate velocity of sound in air.
84. The correct figure that shows, schematically, the wave pattern produced by superposition of two wave of frequencies 9Hz and 11Hz is :



85. 5 beats/ second are heard when a tuning fork is sounded with a sonometer wire under tension, when the length of the sonometer wire is either 0.95m or 1m. The frequency of the fork will be
 (A) 195Hz (B) 251 Hz (C) 150Hz (D) 300Hz
86. A string of length 0.4 m and mass 10^{-2} kg is tightly clamped at its ends. The tension in the string is 1.6N. Identical wave pulses are produced at one end at equal intervals of time, Δt . The minimum value of Δt which allows constructive interference between successive pulses is
 (A) 0.05s (B) 0.10s (C) 0.20s (D) 0.40s
87. Two source of sound S_1 and S_2 produce sound waves of same frequency 660Hz. A listener is moving from source S_1 towards S_2 with a constant speed u m/s and he hears 10 beats/s. The velocity of sound is 330m/s. The u equals:

88. A police car with a siren of frequency 8 kHz is moving with uniform velocity 36 km/hr towards a tall building which reflects the sound waves. The speed of sound in air is 320 m/s. The frequency of the siren heard by the car driver is
 (A) 8.50 kHz (B) 8.25 kHz (C) 7.75 kHz (D) 7.50 kHz
89. A siren placed at a railway platform is emitting sound of frequency 5 kHz. A passenger sitting in a moving train A records a frequency of 5.5 kHz while the train approaches the siren. During his return journey in a different train B he records a frequency of 6.0 kHz while approaching the same siren. The ratio of the velocity of train B to that train A is
 (A) 242/252 (B) 2 (C) 5/6 (D) 11/6
90. A train S1, moving with a uniform velocity of 108 km/h approaches another train S2 standing on a platform. An observer O moves with a uniform velocity of 36 km/h towards S2 as shown in figure. Both the trains are blowing whistles of same frequency 120 Hz. When O is 600 m away from S2 and distance between S1 and S2 is 800 m, the number of beats heard by O is _____.
 [Speed of the sound = 330 m/s]



91. Two men are walking along a horizontal straight line in the same direction. The man in front walks at a speed 1.0 ms^{-1} and the man behind walks at a speed 2.0 ms^{-1} . A third man is standing at a height 12 m above the same horizontal line such that all three men are in a vertical plane. The two walking men are blowing identical whistles which emit a sound of frequency 1430 Hz. The speed of sound in air is 330 ms^{-1} . At the instant, when the moving men are 10 m apart, the stationary man is equidistant from them. The frequency of beats in Hz, heard by the stationary man at this instant, is _____
92. A stationary source emits sound of frequency $f_0 = 492 \text{ Hz}$. The sound is reflected by a large car approaching the source with a speed of 2 ms^{-1} . The reflected signal is received by the source and superposed with the original. What will be the beat frequency of the resulting signal in Hz?
 (Given that the speed of sound in air is 330 ms^{-1} and the car reflects the sound at the frequency it has received.)
93. A student is performing an experiment using a resonance column and a tuning fork of frequency 244 s^{-1} . He is told that the air in the tube has been replaced by another gas (assume that the column remains filled with the gas). If the minimum height at which resonance occurs is $(0.350 \pm 0.005) \text{ m}$, the gas in the tube is

(Useful information: $\sqrt{167RT} = 640 \text{ J}^{1/2} \text{ mole}^{-1/2}$; $\sqrt{140RT} = 590 \text{ J}^{1/2} \text{ mole}^{-1/2}$. The molar masses M in

grams are given there.)

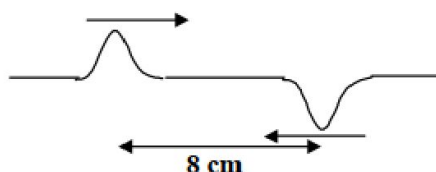
(A) Neon $\left(M = 20, \sqrt{\frac{10}{20}} = \frac{7}{10} \right)$

(B) Nitrogen $\left(M = 28, \sqrt{\frac{10}{28}} = \frac{3}{5} \right)$

(C) Oxygen $\left(M = 32, \sqrt{\frac{10}{32}} = \frac{9}{16} \right)$

(D) Argon $\left(M = 36, \sqrt{\frac{10}{36}} = \frac{17}{32} \right)$

94. Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in the figure. The speed of each pulse is 2cm/s. After 2 seconds, the total energy of the pulses will be

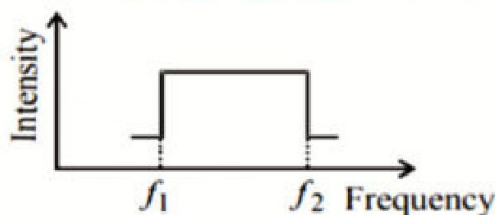


- (A) zero
(B) purely kinetic
(C) purely potential
(D) partly kinetic and partly potential

Passage – 3

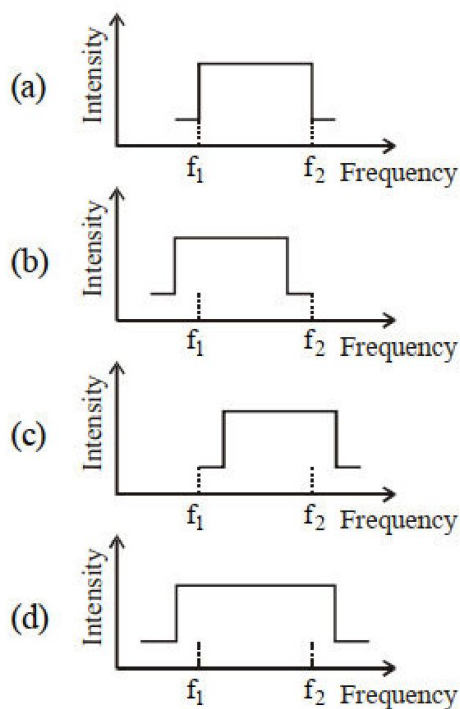
Two trains A and B moving with speed 20m/s and 30m/s respectively in the same direction on the same straight track, with B ahead of A. The engines are at the front ends. The engine of train A blows a long whistle.

Assume that the sound of the whistle is composed of components varying in frequency from $f_1 = 800\text{Hz}$ to $f_2 = 1120\text{Hz}$, as shown in the figure. The spread in the frequency (highest frequency – lowest frequency) is thus 320 Hz. The speed of sound in still air is 340 m/s.



95. The speed of sound of the whistle is
- (A) 340 m/s for passengers in A and 310m/s for passengers in B
(B) 360 m/s for passengers in A and 310 m/s for passengers in B
(C) 310 m/s for passengers in A and 360 m/s for passengers in B
(D) 340 m/s for passengers in both the trains

96. The distribution of the sound intensity of the whistle as observed by the passengers in train A is best represented by



97. The spread of frequency as observed by the passengers in train B is
 (A) 310 Hz (B) 330 Hz (C) 350 Hz (D) 290 Hz

Passage – 4

Waves $y_1 = A \cos(0.5\pi x - 100\pi t)$ and $y_2 = A \cos(0.46\pi x - 92\pi t)$ are travelling along x -axis. (Here x is in m and t is in second)

98. Find the number of times intensity is maximum in time interval of 1 sec.
 (A) 4 (B) 6 (C) 8 (D) 10
99. The wave velocity of louder sound is
 (A) 100m/s (B) 192m/s (C) 200m/s (D) 96m/s
100. The number of times $y_1 + y_2 = 0$ at $x = 0$ in 1 sec is
 (A) 100 (B) 46 (C) 192 (D) 96

KEY & SOL

1. Key: (A)

Sol: Relation between electric field E_0 and magnetic field B_0 of an electromagnetic wave is given by

$$c = \frac{E_0}{B_0} \quad [\text{Here, } c = \text{speed of light}]$$

$$\Rightarrow E_0 = B_0 \times c = 1.25 \times 10^{-7} \times 3 \times 10^8 = 36$$

As the wave is propagating along x – direction, magnetic field is along z – direction and $(\hat{E} \times \hat{B}) \parallel \hat{C}$

$\therefore \vec{E}$ should be along y – direction

So, electric field $\vec{E} = E_0 \sin \vec{E} \cdot (\mathbf{x}, t)$

$$= [-36 \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t) \hat{j}] \frac{\text{V}}{\text{m}}$$

2. Key: (A)

Sol: $\vec{E} = E_0(\hat{x} + \hat{y}) \sin(kz - \omega t)$

Direction of propagation of em wave = $\pm \hat{k}$

Unit vector in the direction of electric field, $\hat{E} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

The direction of electromagnetic wave is perpendicular to both electric and magnetic field.

$$\therefore \hat{k} = \hat{E} \times \hat{B}$$

$$\Rightarrow \hat{k} = \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) \times \hat{B} \Rightarrow \hat{B} = \frac{-\hat{i} + \hat{j}}{\sqrt{2}}$$

$$\therefore \vec{B} = \frac{E_0}{c} (-\hat{x} + \hat{y}) \sin(kz - \omega t)$$

3. Key: (C)

Sol: Relation between electric field and magnetic field for an electromagnetic wave in vacuum is

$$B_0 = \frac{E_0}{c}$$

In free space, its speed $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

Here, μ_0 = absolute permeability, ϵ_0 = absolute permittivity

$$\therefore B_0 = \frac{E_0}{c} = \frac{E_0}{1/\sqrt{\mu_0 \epsilon_0}} = E_0 \sqrt{\mu_0 \epsilon_0}$$

4. Key: (A)

Sol: Electromagnetic wave will propagate perpendicular to the direction of Electric and Magnetic fields

$$\hat{C} = \hat{E} \times \hat{B}$$

Here unit vector $\hat{C}$ is perpendicular to both $\hat{E}$ and $\hat{B}$

Given, $\vec{E} = \hat{k}, \vec{B} = 2\hat{i} - 2\hat{j}$

$$\therefore \hat{C} = \hat{E} \times \hat{B} = \frac{1}{\sqrt{2}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \frac{\hat{i} + \hat{j}}{\sqrt{2}} \Rightarrow \hat{C} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

5. Key: (A)

Sol: Direction of polarisation = $\hat{E} = \hat{k}$

$$\text{Direction of propagation} = \hat{E} \times \hat{B} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

$$\text{But } \vec{E} \cdot \vec{B} = 0 \quad \therefore \hat{B} = \frac{\hat{i} - \hat{j}}{\sqrt{2}}$$

6. Key: (B)

Sol: Given, $\vec{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)$

$$\text{Using, } E_0 = B_0 \times C = 3 \times 10^{-8} \times 3 \times 10^8 = 9 \text{ V/m}$$

$\therefore$ Electric field,

$$\vec{E} = 9 \sin(1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m}$$

7. Key : (C)

Sol: At $t = 0$, $z = \frac{\pi}{k}$

$$\therefore \vec{E} = \frac{E_0}{\sqrt{2}} (\hat{i} + \hat{j}) \cos[\pi] = -\frac{E_0}{\sqrt{2}} (\hat{i} + \hat{j})$$

$$\vec{F}_E = q\vec{E}$$

Force due to electric field will be in the direction $-\frac{(\hat{i} + \hat{j})}{\sqrt{2}}$ force due to magnetic field is in direction

$q(\vec{v} \times \vec{B})$ and $\vec{v} \parallel \vec{k}$. Therefore, it is parallel to $\vec{E}$.

$$\Rightarrow \vec{F}_{\text{net}} = \vec{F}_E + \vec{F}_B \text{ is antiparallel to } \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

8. Key: (B)

$$\text{Sol: } B_0 = \frac{E_0}{C} = \frac{60}{3 \times 10^8}$$

$$= 20 \times 10^{-8} \text{ T} = 2 \times 10^{-7} \text{ T}$$

$$K = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi \times 23.9 \times 10^9}{3 \times 10^8} = 500$$

$$\text{Therefore, } \vec{B} = B_0 \sin(kz - \omega t)$$

$$= 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$$

9. Key: (A)

$$\text{Sol: } \frac{E_0}{B_0} = C \Rightarrow B_0 = \frac{E_0}{C}$$

$$\text{Given that } \vec{E} = E_0 \cos(kz) \cos(\omega t) \hat{i}$$

$$\vec{E} = \frac{E_0}{2} [\cos(kz - \omega t) \hat{i} - \cos(kz + \omega t) \hat{i}]$$

Correspondingly

$$\vec{B} = \frac{B_0}{2} [\cos(kz - \omega t) \hat{j} - \cos(kz + \omega t) \hat{j}]$$

$$\vec{B} = \frac{B_0}{2} \times 2 \sin kz \sin \omega t$$

$$\vec{B} = \left(\frac{E_0}{C} \sin kz \sin \omega t \right) \hat{j}$$

10. Key: (B)

$$\begin{aligned} \text{Sol: } F &= (1+r) \frac{IA}{C} \\ &= \frac{(1+0.25) \times 50 \times 1}{3 \times 10^8} = 20 \times 10^{-8} \text{ N} \end{aligned}$$

11. Key: (C)

$$\text{Sol: } E_0 = cB_0 = 3 \times 10^8 \times 1.6 \times 10^{-6} = 4.8 \times 10^2 \text{ V/m}$$

$$\text{Also } \vec{S} \Rightarrow \vec{E} \times \vec{B}$$

$$\text{Or } -\vec{K} \Rightarrow \vec{E} \times (2\hat{i} + \hat{j})$$

$$\text{Therefore direction of } \vec{E} \times (2\hat{i} + \hat{j})$$

12. Key: (D)

Sol: EM wave intensity

$$\Rightarrow I = \frac{\text{Power}}{\text{Area}} = \frac{1}{2} \epsilon_0 E_0^2 c$$

[where E_0 = maximum electric field]

$$\Rightarrow \frac{27 \times 10^{-3}}{10 \times 10^{-6}} = \frac{1}{2} \times 9 \times 10^{-12} \times E_0^2 \times 3 \times 10^8$$

$$\Rightarrow E_0 = \sqrt{2} \times 10^3 \text{ kV/m} = 1.4 \text{ kV/m}$$

13. Key: (D)

$$\text{Sol: } \vec{E} = 10 \cos(6x + 8z + ct) \hat{j}$$

$$\text{Now } \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \Rightarrow \vec{B} = -\int \vec{\nabla} \times \vec{E} dt$$

$$\vec{B} = \cos(6x + 8z + 10ct) [8\hat{i} - 6\hat{k}]$$

14. Key: (B)

Sol: $\hat{E} \times \hat{B}$ should give the direction of wave propagation

15. Key: (C)

Sol: Given, Electric field component of monochromatic radiation,

$$(\vec{E}) = 2E_0 \hat{i} \cos kz \cos \omega t$$

$$\text{We know that, } \frac{dE}{dz} = -\frac{dB}{dt}$$

$$\frac{dE}{dz} = -2E_0 k \sin kz \cos \omega t = -\frac{dB}{dt}$$

$$dB = +2E_0 k \sin kz \cos \omega t dt \quad \dots (i)$$

Integrating eqⁿ.(i), we have

$$B = +2E_0 k \sin kz \int \cos \omega t dt$$

Magnetic field is given by,

$$= +2E_0 \frac{k}{\omega} \sin kz \sin \omega t$$

We also know that,

$$\frac{E_0}{B_0} = \frac{\omega}{k} = c$$

Magnetic field vector,

$$\vec{B} = \frac{2E_0}{c} \hat{j} \sin kz \sin \omega t$$

16. Key: (B)

Sol: Wavelength of monochromatic green light

$$= 5.5 \times 10^{-5} \text{ cm}$$

$$\text{Intensity } I = \frac{\text{Power}}{\text{Area}} = \frac{100 \times (3 \times 100)}{4\pi(5)^2} = \frac{3}{100\pi} \text{ Wm}^{-2}$$

Now, half of this intensity (I) belong to electric field and half of that to magnetic field, therefore,

$$\frac{I}{2} = \frac{1}{4} \epsilon_0 E_0^2 C$$

17. Key: (C)

Sol: $\frac{E, \text{Decreases}}{\gamma - \text{rays} \quad X - \text{rays} \quad \text{uv} - \text{rays} \quad \text{Visible rays} \quad \text{IR rays} \quad \text{Radio}}$

VIBGYOR Microwaves waves

Radio wave < yellow light < blue light < X - rays

(Increasing order of energy)

18. Key: (C)

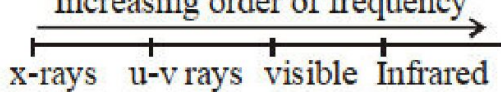
Sol: Gamma rays < X - rays < Ultra violet < Visible rays < Infrared rays < Microwaves < Radio waves.

20. Key: (A)

$$\text{Sol: } E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$$

$$\Rightarrow \lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{11 \times 1000 \times 1.6 \times 10^{-19}}$$

$$= 12.4 \text{ \AA}$$

Increasing order of frequency


Wavelength range of visible region is 4000 Å to 7800 Å

21. Key: D

22. Key: (C)

Sol: Optical Fibre Communication – Infrared light

Radar – Radio Waves

Sonar – Ultrasound

Mobile Phones – Microwaves

23. Key: (D)

Sol: (1) Infrared rays are used to treat muscular strain because these are heat rays.

(2) Radio waves are used for broadcasting because these waves have very long wavelength ranging from few centimeters to few hundred kilometres.

(3) X - rays are used to detect fracture of bones because they have high penetrating power but they have high penetrating power but they can't penetrate through denser medium like bones.

(4) Ultraviolet rays are absorbed by ozone of the atmosphere.

24. Key: (A)

Sol: Wavelength emitted by atomic hydrogen in interstellar space – part of short radio wave or microwave of electromagnetic spectrum Doublet of sodium – visible radiation.

25. Key: (D)

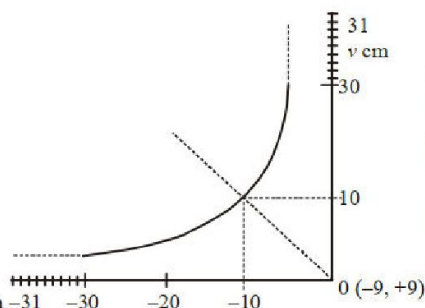
Sol: Vibration of atoms and molecules 700 nm to 1 mm Radioactive decay of the nucleus $< 10^{-3}$ nm
Magnetron valve 1 mm to 0.1 m.

26. Key: (C)

Sol: From the graph $u = -10\text{cm}$; $v = 10\text{cm}$ $\Delta u = \Delta v = 0.1$

$$\text{From lens formula, } \frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{10} - \frac{1}{-10}$$

$$\therefore f = 5\text{cm}$$



Differentiating lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{\Delta f}{f^2} = \frac{\Delta v}{v^2} + \frac{\Delta u}{u^2} \quad (\text{for maximum error in } f)$$

$$\Rightarrow \frac{\Delta f}{25} = \frac{0.1}{(10)^2} + \frac{0.1}{(10)^2}$$

$$\Rightarrow \Delta f = 25 \times 0.1 \times 2 \times 0.01 = 0.05$$

$$\therefore \text{Focal length, } f \pm \Delta f = (5.00 \pm 0.05)\text{cm}$$

27. Key: (C)

Sol: Angle of deviation for the first prism P_1

$$\delta_1 = (\mu_1 - 1)A_1$$

Angle of deviation for the second prism P_2

$$\delta_2 = (\mu_2 - 1)A_2$$

For dispersion without deviation, $\delta_{\text{net}} = 0$

$$\therefore (\mu_1 - 1)A_1 = (\mu_2 - 1)A_2 \text{ or, } A_2 = \frac{(1.54 - 1)}{(1.72 - 1)} 4^\circ = 3^\circ$$

28. Key: (D)

Sol: $e \rightarrow f$ when the ray enters from the rectangular block to prism then angle of incidence $>$ angle of refraction, so $\mu_2 > \mu_1$. The ray then moves away from the normal when it emerges out of the rectangular block. Therefore $\mu_2 > \mu_3$.

$e \rightarrow g$. As there is no deviation of the ray as it emerges out of the prism, $\therefore \mu_2 = \mu_1$.

$e \rightarrow h$. As the ray emerges out of prism, it moves away from the normal. $\therefore \mu_2 < \mu_1$.

And the ray moves away from the normal as it emerges out of the rectangular block,

$$\therefore \mu_2 > \mu_3$$

$e \rightarrow i$. At the prism surface, total internal reflection has taken place.

$$\therefore \text{Critical angle } 45^\circ > C \Rightarrow \sin 45^\circ > \sin C$$

$$\Rightarrow \frac{1}{\sqrt{2}} > \frac{\mu_2}{\mu_1} \quad \therefore \mu_1 > \sqrt{2}\mu_2$$

29. Key: (D)

Sol: For telescope

$$\text{Tube length (L)} = f_o + f_e = 60$$

$$\text{And magnification (m)} = \frac{f_o}{f_e} = 5 \Rightarrow f_o = 5f_e$$

$$\therefore f_o = 50\text{cm and } f_e = 10\text{cm}$$

Hence focal length of eye – piece, $f_e = 10\text{cm}$

30. Key: (A)

Sol: According question, $M = 375$

$$L = 150\text{ mm, } f_o = 5\text{mm and } f_e = ?$$

$$\text{Using, magnification, } M \approx \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$$

$$\Rightarrow 375 = \frac{150}{5} \left(1 + \frac{250}{f_e} \right) \quad (\because D = 25\text{cm} = 250\text{mm})$$

$$\Rightarrow 12.5 = 1 + \frac{250}{f_e} \Rightarrow f_e = \frac{250}{11.5} = 21.7 \approx 22\text{mm}$$

31. Key: (B)

Sol: A telescope magnifies by making the object appearing closer.

32. To determine refractive index of glass slab using a travelling microscope, minimum number of readings required are:

(A) Two

(B) Four

(C) Three

(D) Five

Key: (C)

Sol: Reading one P without slab

Reading two P with slab

Reading three P with saw dust

Minimum three readings are required to determine refractive index of glass slab using a travelling microscope.

33. Key: (C)

Sol: Magnifying power of telescope,

$$MP = \frac{\beta(\text{angle subtended by image at eye piece})}{\alpha(\text{angle subtended by object on objective})}$$

$$\text{Also, } MP = \frac{f_o}{f_e} = \frac{150}{5} = 30$$

$$\alpha = \frac{50}{1000} = \frac{1}{20} \text{ rad}$$

$$\therefore \beta = \theta = MP \times \alpha = 30 \times \frac{1}{20} = \frac{3}{2} = 1.5 \text{ rad}^{-1}$$

$$\text{Or, } \beta = 1.5 \times \frac{180^\circ}{\pi} \approx 84^\circ$$

34. Key: (A)

Sol: Given $f_o = 1.2\text{cm; } f_e = 3.0\text{cm}$

$$u_0 = 1.25\text{cm}; M_\infty = ?$$

$$\text{From } \frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{v_0} \Rightarrow \frac{1}{1.2} = \frac{1}{v_0} - \frac{1}{(-1.25)}$$

$$\Rightarrow \frac{1}{v_0} = \frac{1}{1.2} - \frac{1}{1.25} \Rightarrow v_0 = 30\text{cm}$$

Magnification at infinity,

$$M_\infty = -\frac{v_0}{u_0} \times \frac{D}{f_e} = \frac{30}{1.25} \times \frac{25}{3}$$

($\because D = 25\text{cm}$ least distance of distinct vision) = 200

Hence the magnifying power of the compound microscope is 200

35. Key: (A)

Sol: Here $f_0 = 2\text{cm}$ and $f_e = 3\text{cm}$.

Using lens formula for eye piece

$$\Rightarrow \frac{-1}{u_e} + \frac{1}{\infty} = \frac{1}{3} \Rightarrow u_e = -3\text{cm}$$

[$\because$ final image formed by the eyepiece at ∞]

Given distance between objective and eye piece = 15 cm

$\therefore$ Distance of image formed by the objective

$$V_0 = 15 - 3 = 12\text{cm}$$

Let u_0 be the object distance from objective, then for objective lens

$$\Rightarrow \frac{-1}{u_0} = \frac{1}{2} - \frac{1}{12} = \frac{5}{12} \quad \therefore u_0 = -\frac{12}{5} = -2.4\text{cm}$$

36. Key: 50

Sol: Given: length of compound microscope, $L = 10\text{ cm}$

Focal length of objective $f_0 = 1\text{cm}$ and of eye – piece,

$$f_e = 5\text{cm}$$

$$u_0 = f_e = 5\text{cm}$$

Final image formed at infinity (∞), $v_e = \infty$

$$v_0 = 10 - 5 = 5$$

Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

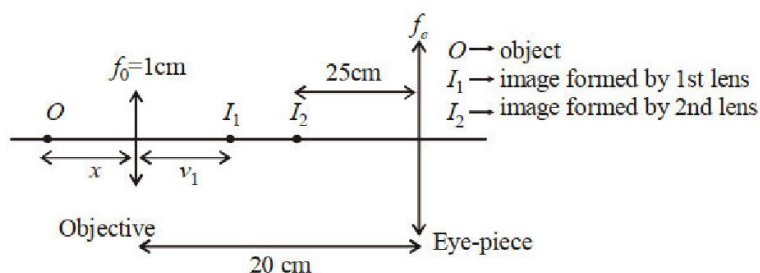
$$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0} \Rightarrow \frac{1}{5} - \frac{1}{u_0} = \frac{1}{1} \Rightarrow u_0 = -\frac{5}{4}\text{cm}$$

$$\text{Or, } \frac{5}{4} = \frac{N}{40} \quad \therefore N = \frac{200}{4} = 50\text{cm}$$

37. Key: 4.48

Sol: According to question, final image i.e. $v_2 = 25\text{cm}$

$$f_0 = 1\text{cm}; \text{ magnification, } m = m_1 m_2 = 100$$



Using lens formula,

$$\text{For first lens or objective} = \frac{1}{v_1} - \frac{1}{-x} = \frac{1}{f_o} \Rightarrow v_1 = \frac{x}{x-1}$$

$$\text{Also magnification } |m_1| = \left| \frac{v_1}{u_1} \right| = \frac{1}{x-1}$$

For 2nd lens or eye - piece, this is acting as object

$$\therefore u_2 = -(20 - v_1) = -\left(20 - \frac{x}{x-1}\right) \text{ and } v_2 = -25\text{cm}$$

$$\text{Angular magnification } |m_A| = \left| \frac{D}{u_2} \right| = \frac{25}{|u_2|}$$

$$\text{Total magnification } m = m_1 m_A = 100$$

$$\left(\frac{1}{x-1} \right) \left(\frac{25}{20 - \frac{x}{x-1}} \right) = 100$$

$$\Rightarrow \frac{25}{20(x-1) - x} = 100 \Rightarrow 1 = 80(x-1) - 4x$$

$$\Rightarrow 76x = 81 \Rightarrow x = \frac{81}{76}$$

$$\Rightarrow u_2 = -\left(20 - \frac{\frac{81}{76}}{\frac{81}{76}-1}\right) = \frac{-19}{5}$$

Again, using lens formula for eye- piece

$$\frac{1}{-25} - \frac{1}{\frac{-19}{5}} = \frac{1}{f_e} = \frac{25 \times 19}{106} \approx 4.48\text{cm}$$

38. Key: ABCD

Sol: In an astronomical telescope the distance between the objective lens and eyepiece lens

$$L = f_o + f_e = 16 + 0.02 = 16.02\text{m}$$

$$\text{Angular magnification} = -\frac{f_{\text{objective}}}{f_{\text{eyepiece}}} = \frac{-16}{0.02} = -800$$

The image seen by the astronomical telescope is inverted. Also the objective is larger than eye piece.

39. Key: (D)

Sol: In an astronomical telescope when the final image is formed at infinity :

Angular magnification

$$\therefore M = \frac{f_0}{f_e} = 5 \Rightarrow f_0 = 5f_e$$

And the separation between the objective and eye – piece.

$$L = f_0 + f_e = 36 \Rightarrow 5f_e + f_e = 36 \text{ or } f_e = 6\text{cm}$$

$$\therefore f_0 = 5f_e = 5 \times 6 = 30\text{cm}$$

40. Key: (C)

Sol: From graph equation of SHM

$$X = A \cos \omega t$$

(1) At $\frac{3T}{4}$ particle is at mean position

$$\therefore \text{Acceleration} = 0, \text{Force} = 0$$

(2) At T particle again at extreme position so acceleration is maximum.

(3) At $t = \frac{T}{4}$, particle is at mean position so velocity is maximum.

$$\text{Acceleration} = 0$$

(4) When KE = PE

$$\Rightarrow \frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2$$

Here, A = amplitude of SHM

x = displacement from mean position

$$\Rightarrow A^2 = 2x^2 \Rightarrow x = \frac{+A}{\sqrt{2}}$$

$$\Rightarrow A^2 = A^2 \cos^2 \omega t \Rightarrow t = \frac{T}{2}$$

$\therefore x = -A$ which is not possible

$\therefore 1, 2$ and 3 are correct.

41. Key: (B)

Sol: We know that $V = \omega \sqrt{A^2 - x^2}$

$$\text{Initially } V = \omega \sqrt{A^2 - \left(\frac{2A}{3}\right)^2}$$

$$\text{Finally } 3V = \omega \sqrt{A'^2 - \left(\frac{2A}{3}\right)^2}$$

Where A' = final amplitude (Given at $x = \frac{2A}{3}$, velocity to trebled)

On dividing we get

$$\frac{3}{1} = \frac{\sqrt{A'^2 - \left(\frac{2A}{3}\right)^2}}{\sqrt{A^2 - \left(\frac{2A}{3}\right)^2}}$$

$$9 \left[A^2 - \frac{4A^2}{9} \right] = A'^2 - \frac{4A^2}{9} \quad \therefore A' = \frac{7A}{3}$$

42. Key: 0.06m

Sol: $m = 0.2 \text{ kg}$ $f = \frac{25}{\pi} \text{ Hz}$

P.E. = $4/9$ T.E

$$\frac{1}{2} kx^2 = \frac{4}{9} \times \frac{1}{2} kA^2; x = \frac{2}{3} A$$

$$A = \frac{3}{2} x = \frac{3}{2} \times 0.04$$

$$A = 0.06 \text{ m}$$

43. Key: (B)

Sol: From graph it is clear that when

$$L = 1 \text{ m}, T^2 = 4 \text{ s}^2$$

As we know,

$$T = 2\pi \sqrt{\frac{L}{g}} \Rightarrow g = \frac{4\pi^2 L}{T^2}$$

$$= 4 \times \left(\frac{22}{7}\right)^2 \times \frac{1}{4} = \left(\frac{22}{7}\right)^2$$

$$\therefore g = \frac{484}{49} = 9.87 \text{ m/s}^2$$

44. Key: (C)

Sol: Mass of bigger body $M = 4 \text{ kg}$

Mass of smaller body $m = 1 \text{ kg}$

Smaller mass ($m = 1 \text{ kg}$) executes S.H.M. of angular frequency $\omega = 25 \text{ rad/s}$

Amplitude $x = 1.6 \text{ cm} = 1.6 \times 10^{-2}$

As we know,

$$T = 2\pi \sqrt{\frac{m}{K}} \text{ or } \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{K}}$$

$$\text{Or, } \frac{1}{25} = \sqrt{\frac{1}{K}} \quad [\because m = 1 \text{ kg}; \omega = 25 \text{ rad/s}]$$

$$\text{Or } K = 625 \text{ Nm}^{-1}$$

The maximum force exerted by the system on the floor

$$= Mg + Kx + mg$$

$$= 4 \times 10 + 625 \times 1.6 \times 10^{-2} + 1 \times 10 = 40 + 10 + 10 = 60 \text{ N}$$

45. Key: (C)

Sol: $\frac{Mg}{A} = P_0$

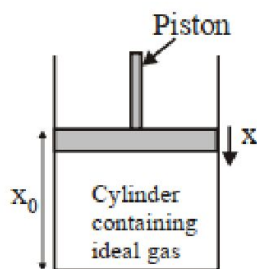
$$P_0 V_0^\gamma = PV^\gamma$$

$$Mg = P_0 A \quad \dots (1)$$

Let piston is displaced by distance x

$$P_0 A x_0^\gamma = PA(x_0 - x)^\gamma$$

$$P = \frac{P_0 x_0^\gamma}{(x_0 - x)^\gamma}$$



$$Mg - \left(\frac{P_0 x_0^\gamma}{(x_0 - x)^\gamma} \right) A = F_{\text{restoring}}$$

$$P_0 A \left(1 - \frac{x_0^\gamma}{(x_0 - x)^\gamma} \right) = F_{\text{restoring}} [x_0 - x \approx x_0]$$

$$F = - \frac{\gamma P_0 A x}{x_0}$$

$\therefore$ Frequency with which piston executes SHM.

$$f = \frac{1}{2\pi} \sqrt{\frac{\gamma P_0 A}{x_0 M}} = \frac{1}{2\pi} \sqrt{\frac{\gamma P_0 A^2}{M V_0}}$$

46. Key: (A)

Sol: Let T_1 and T_2 be the time period of the two pendulums

$$T_1 = 2\pi \sqrt{\frac{l}{g}} \quad \text{and} \quad T_2 = 2\pi \sqrt{\frac{4}{g}}$$

As $l_1 < l_2$ therefore $T_1 < T_2$

Let at $t = 0$ they start swinging together. Since their time period are different, the swinging will not be in unison always. Only when number of completed oscillations differ by an integer, the two pendulums will again begin to swing together. Let longer length pendulum complete n oscillation and shorter length pendulum complete $(n+1)$ oscillation. For unison swinging

$$(n+1)T_1 = nT_2$$

$$(n+1) \times 2\pi \sqrt{\frac{l}{g}} = (n) \times 2\pi \sqrt{\frac{4}{g}}$$

$$\Rightarrow n = 1$$

$$\therefore n+1 = 1+1 = 2$$

47. Key: (B)

Sol: Given : $y = kt^2$

$$\therefore \frac{dy}{dt} = v = 2kt \quad \text{and} \quad \frac{d^2y}{dt^2} = a = 2k$$

$$\text{Or } a_y = 2m/s^2 \quad (\because k = 1m/s^2 \text{ given})$$

$$\text{We know that } T = 2\pi \sqrt{\frac{\ell}{g}} \quad \text{or, } T_1 = 2\pi \sqrt{\frac{1}{g}}$$

$$T_2 = 2\pi \sqrt{\frac{1}{g+a_y}}$$

$$\therefore \frac{T_1^2}{T_2^2} = \frac{g+a_y}{g} \Rightarrow \frac{T_1^2}{T_2^2} = \frac{12}{10} = \frac{6}{5}$$

48. Key: (A)

Sol: Here acceleration is same for both P and Q as block Q oscillates but does not slip. The $(P-Q)$ system oscillates with angular frequency ω . The spring is stretched by A .

Angular frequency of the system,

$$\omega = \sqrt{\frac{k}{m+m}} = \sqrt{\frac{k}{2m}}$$

Maximum acceleration of the system is SHM

$$a_{\max} = A\omega^2 = A\sqrt{\left(\frac{k}{2m}\right)^2} = \frac{KA}{2m}$$

This acceleration to the lower block is provided by friction

$\therefore$ Maximum force of friction

$$f_{\max} = ma_{\max} = m\left(\frac{kA}{2m}\right) = \frac{kA}{2}$$

49. Key: (A)

Sol: Given: $U(x) = k|x|^3$

$$\therefore [k] = \frac{[U]}{[x^3]} = \frac{ML^2T^{-2}}{L^3} = ML^{-1}T^{-2}$$

Now time period may depend on $T \propto (\text{mass})^x$

$(\text{amplitude})^y (k)^z$

$$\therefore [M^0L^0T] = [M]^x [L]^y [ML^{-1}T^{-2}]^z = [M^{x+z}L^{y-z}T^{-2z}]$$

Equating the powers, we get

$$-2z = 1 \text{ or } z = -1/2$$

$$y - z = 0 \text{ or } y = z = -1/2$$

$$\text{Hence } T \propto (\text{amplitude})^{-1/2} \propto a^{-1/2}$$

$$\text{Therefore, } T \propto \frac{1}{\sqrt{a}}$$

50. Key: (D)

Sol: Here, $A = L^2$ and $\theta = \frac{x}{L}$

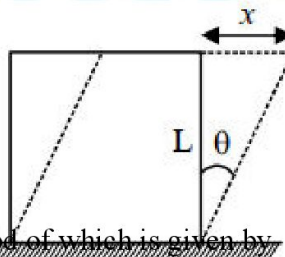
Modulus of rigidity, $\eta = F / A \theta$

$\therefore$ Restoring force

$$F = -\eta A \theta = -\eta L x$$

Or, acceleration,

$$a = \frac{F}{M} = -\frac{\eta L}{M} x$$



Since, $a \propto -x$ So, oscillations are SHM. Therefore, time period of which is given by

$$T = 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}} = 2\pi \sqrt{\frac{x}{a}}$$

$$\text{Or, } T = 2\pi \sqrt{\frac{M}{\eta L}}$$

51. Key: (B)

Sol: If x is the displacement then,

Net restoring force = extra upthrust + extra spring force

$$\therefore M\omega^2 x = [\rho Ag + k]x$$

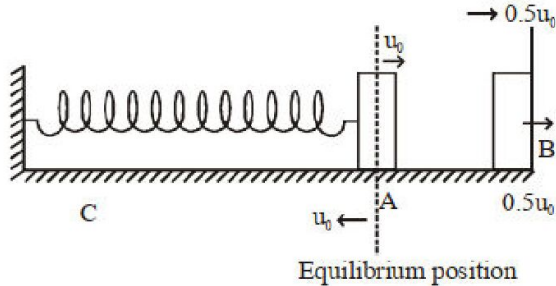
$$\Rightarrow \omega = \left[\frac{\rho Ag + k}{M} \right]^{1/2} \Rightarrow v = \frac{1}{2\pi} \left[\frac{\rho Ag + k}{M} \right]^{1/2}$$

52. Key: (AD)

Sol: The particle collides elastically with rigid wall.

$$\therefore e = \frac{V}{0.5u_0} = 1 \Rightarrow V = 0.5u_0$$

i.e., the particle rebounds with the same speed. Therefore the particle will return to its equilibrium position with speed u_0 .



The velocity of the particle becomes $0.5u_0$ after time t . Using, equation $V = V_{\max} \cos \omega t$

$$0.5u_0 = u_0 \cos \omega t$$

$$\therefore \frac{\pi}{3} = \frac{2\pi}{t} \times T \Rightarrow t = \frac{T}{6}$$

$$\text{The time period } T = 2\pi\sqrt{\frac{m}{k}} \quad \therefore t = \frac{\pi}{3}\sqrt{\frac{m}{k}}$$

$$\text{The time taken by the particle to pass through the equilibrium for the first time} = 2t = \frac{2\pi}{3}\sqrt{\frac{m}{k}}$$

The time taken for the maximum compression

$$\begin{aligned} &= t_{AB} + t_{BA} + t_{AC} \\ &= \frac{\pi}{3}\sqrt{\frac{m}{k}} + \frac{\pi}{3}\sqrt{\frac{m}{k}} + \frac{\pi}{3}\sqrt{\frac{m}{k}} = \pi\sqrt{\frac{m}{k}} \left[\frac{1}{3} + \frac{1}{3} + \frac{1}{3} \right] \\ &= \frac{7\pi}{6}\sqrt{\frac{m}{k}} \end{aligned}$$

The time taken for particle to pass through the equilibrium position second time

$$= 2 \left[\frac{\pi}{3}\sqrt{\frac{m}{k}} \right] + \pi\sqrt{\frac{m}{k}} = \pi\sqrt{\frac{m}{k}} \left(\frac{2}{3} + 1 \right) = \frac{5\pi}{3}\sqrt{\frac{m}{k}}$$

53. Key: (C)

Sol: Time of half the amplitude is $= 2s$

Using, $A = A_0 e^{-kt}$

$$\frac{A_0}{2} = A_0 e^{-k \times 2} \quad \dots (i)$$

$$\text{And } \frac{A_0}{1000} = A_0 e^{-kt} \quad \dots (ii)$$

Dividing (i) by (ii) and solving, we get

$$t = 20s$$

54. Key: (B)

Sol: In first collision mu momentum will be imparted to system, in second collision when momentum of $(M + m)$ is in opposite direction mu momentum of particle will make its momentum zero.

On 13th collision, $m \rightarrow M+12$; $M+13m \rightarrow V$

$$mu = (M+13m)v \Rightarrow v = \frac{mu}{M+13m} = \frac{u}{15}$$

$$v = \omega A \Rightarrow \frac{u}{15} = \sqrt{\frac{K}{M+13m}} \times A$$

Putting value of M , m , u and K we get amplitude

$$A = \frac{1}{15} \sqrt{\frac{75}{1}} = \frac{1}{\sqrt{3}}$$

55. Key: (B)

Sol: The change in time period compared to the undamped oscillator increase by 8%

$$\omega = \sqrt{\frac{k}{m} - \frac{r^2}{4m^2}} = \omega_0 \sqrt{1 - \frac{r^2}{4mk}} \approx \omega_0 \left(\frac{1-r^2}{8mk} \right)$$

$$\Rightarrow \omega - \omega_0 = -\omega_0 \frac{r^2}{8mk}$$

$$\Rightarrow \frac{\Delta\omega}{\omega_0} = \frac{-r^2}{8mk} = -8\% \Rightarrow \frac{\Delta T}{T_0} = +8\%$$

56. Key: (C)

Sol: $\because A = A_0 e^{-bt/2m}$

(where, A_0 = maximum amplitude)

According to the questions, after 5 second,

$$0.9A_0 = A_0 e^{-\frac{b(5)}{2m}} \quad \dots (i)$$

After 10 more second,

$$A = A_0 e^{-\frac{b(15)}{2m}} \quad \dots (ii)$$

From eqⁿs (i) and (ii)

$$A = 0.729A_0$$

$$\therefore \alpha = 0.729$$

57. Key: 4

Sol: As we know, $Y = \frac{FL}{Al} \Rightarrow F = \left(\frac{YA}{L} \right) l$

Also, $F = Kl$

$$\text{Or, } Kl = \left(\frac{YA}{L} \right) l \Rightarrow K = \frac{YA}{L}$$

$$\text{Angular frequency } \omega = \sqrt{\frac{K}{m}} \text{ or, } \omega = \sqrt{\frac{YA}{ml}}$$

$$\text{Or, } 140 = \sqrt{\frac{n \times 10^9 \times 4.9 \times 10^{-7}}{0.1 \times 1}} \quad (\because Y = 9 \times 10^9 \text{ given})$$

$$\therefore n = 4$$

58. Key: BD

Sol: For first harmonic oscillator,

Mass = m

Angular frequency = ω_1

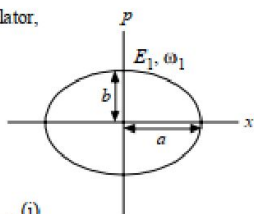
Amplitude = a

Total energy = E_1

Maximum momentum,

$$p_{\max} = b$$

$$E_1 = \frac{1}{2} m \omega_1^2 a^2$$



Passage – 1 (59-61)

Key: D

Sol: When the ball is thrown upwards, at the point of throw (O) the linear momentum is in upwards direction (and has a maximum value) and the position is zero. As the time passes, the ball moves upwards and its momentum goes on decreasing and the position becomes positive. The momentum becomes zero at the topmost point.

As the time increases, the ball starts moving down with an increasing linear momentum in the downward direction (negative) and reaches back to its original position.

60. Key: (C)

Sol: In SHM mechanical energy, $E_2 \propto (\text{amplitude})^2$

$$\therefore E \propto (2a)^2 \text{ \& } E_2 \propto a^2$$

$$\therefore \frac{E_1}{E_2} = 4$$

61. Key: (B)

Sol: When the position of the mass is at one extreme end in the positive side (the topmost point), the momentum is zero. As the mass moves towards the mean position the momentum increase in the negative direction.

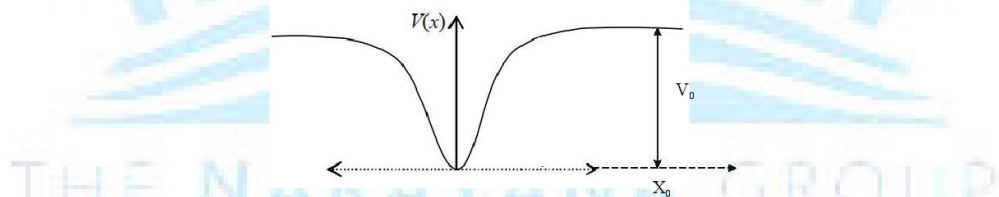
As the mass is oscillating in water its amplitude will go on decreasing and the amplitude will decrease with time.

Passage - 2

When a particle of mass m moves on the x – axis in a potential of the form $V(x) = kx^2$ it performs

simple harmonic motion. The corresponding time period is proportional to $\sqrt{\frac{m}{k}}$, as can be seen

easily using dimensional analysis. However, the motion of a particle can be periodic even when its potential energy increase on both sides of $x = 0$ in a way different from kx^2 and its total energy is such that the particle does not escape to infinity. Consider a particle of mass m moving on the x – axis. Its potential energy is $V(x) = \alpha x^4$ ($\alpha > 0$) for $|x|$ near the origin and becomes a constant equal to V_0 for $|x| \geq X_0$ (see figure).



62. Key: (C)

Sol: The particle will not perform oscillations if energy ≤ 0 . Therefore $E > 0$. If $E = V_0$, the potential energy will become constant as depicted in the graph given. In this case also the particle will not oscillate.

$$\therefore E > V_0$$

$$\therefore V_0 > E > 0$$

63. Key: B

Sol: Potential energy, $V = \alpha x^4$ given

$$\therefore \alpha = \frac{\text{Potential energy}}{x^4} = \frac{ML^2T^{-2}}{L^4} = [ML^{-2}T^{-2}]$$

$$\text{Now } \frac{1}{A} \sqrt{\frac{m}{a}} = \frac{1}{L} \sqrt{\frac{M}{ML^{-2}T^{-2}}} = T$$

64. Key: (D)

Sol: $F = \frac{-dV(x)}{dx}$

$\therefore$ As $V(x) = \text{constant}$ for $x > X_0$

$\therefore F = 0$ and hence $a = 0$ for $x > X_0$

65. Key: (A)

Sol: Polariser A and B have same alignment of transmission axis.

Let assume polariser c is introduced at θ angle

$$\frac{1}{2} \cos^2 \theta \times \cos^2 \theta = \frac{1}{3}$$

$$\text{Or, } \cos^4 \theta = \frac{2}{3} \Rightarrow \cos \theta = \left(\frac{2}{3}\right)^{1/4}$$

66. Key: A

Sol: Use relativistic doppler's effect as velocity of observer is not small as compared to light

$$f = f_0 \sqrt{\frac{c+v}{c-v}}; V = \text{relative speed of approach}$$

$$f_0 = 10\text{GHz}$$

$$f = 10 \sqrt{\frac{c + \frac{c}{2}}{c - \frac{c}{2}}} = 10\sqrt{3} = 17.3\text{GHz}$$

67. Key: (D)

Sol: According to malus law, intensity of emerging beam is given by,

$$I = I_0 \cos^2 \theta$$

$$\text{Now, } I_{A'} = I_A \cos^2 30^\circ$$

$$I_{B'} = I_B \cos^2 60^\circ$$

$$\text{As } I_{A'} = I_{B'}$$

$$\Rightarrow I_A \times \frac{3}{4} = I_B \times \frac{1}{4}; \frac{I_A}{I_B} = \frac{1}{3}$$

68. Key: (C)

Sol: Here, $\sin \theta_{ic} / \sin \theta_{ib} = 1.28$

As we know,

$$\mu = \frac{\sin \theta_{ib}}{\sin \left(\frac{\pi}{2} - \theta_{ib} \right)}$$

Where, θ_{ib} is Brewster's angle of incidence,

$$\text{And, } \mu = \frac{1}{\sin \theta_{ic}}$$

On solving we get, relative refractive index of the two media.

69. Key: (B)

Sol: In a single slit experiment, for diffraction maxima,

$$a \sin \theta = n\lambda$$

According to question,

$$(2 \times 1 + 1) \frac{\lambda}{2} = 1 \times 6600$$

$$\left(\because \lambda_R = 6600 \text{ \AA} \right)$$

$$\lambda = \frac{6600 \times 2}{3}$$

$$\lambda = 4400 \text{ \AA}$$

70. Key: (B)

71. Key: (D)

Sol: For diffraction pattern to be observed, the width of slit should be comparable to the wave length of light used. The wavelength of X – rays ($1-100 \text{ \AA}$) $\ll$ slit width 0.6 mm .

72. Key: (C)

Sol: At the angular position of the first diffraction minimum, the phase difference (θ) between the wavelets from the opposite edges of the slit = 2π radian.

73. Key: (D)

Sol: The distance between the first dark fringe on either side of the central maximum = width of central maximum. Hence distance between two dark fringes on either side

$$= \frac{2D\lambda}{a} = \frac{2 \times 2 \times 600 \times 10^{-9}}{10^{-3}} = 2.4 \times 10^{-3} \text{ m} = 2.4 \text{ mm}$$

74. Key: (D)

Sol: For first minima the path difference between the rays coming from the two edges should be λ therefore corresponding phase difference

$$\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \times \lambda = 2\pi$$

75. Key: 198

Sol: For obtaining secondary minima at a point path difference should be integral multiple of wavelength

$$\therefore d \sin \theta = n\lambda$$

$$\therefore \sin \theta = \frac{n\lambda}{d}$$

For n to be maximum $\sin \theta = 1$

$$n = \frac{d}{\lambda} = \frac{6 \times 10^{-5}}{6 \times 10^{-7}} = 100$$

Total number of minima on one side = 99

Total number of minima = 198.

76. Key: $a \rightarrow p; b \rightarrow r; c \rightarrow r; d \rightarrow p, q, r, s$

Sol: (A) More the radius of aperture more is the amount of light entering the telescope.

$$(B) \text{ Angular magnification, } M = \frac{f_o}{f_e}$$

$$(C) \text{ Length of telescope, } L = f_o + f_e$$

Sharpness of image depends on dispersion of lens, spherical aberration and radius of aperture.

77. Key: (D)

Sol: Total length of the wire; $L = 114\text{cm}$

$$n_1 : n_2 : n_3 = 1 : 3 : 4$$

Let L_1, L_2 and L_3 be the lengths of the three parts

$$\text{As } n \propto \frac{1}{L}$$

$$\therefore L_1 : L_2 : L_3 = \frac{1}{1} : \frac{1}{3} : \frac{1}{4} = 12 : 4 : 3$$

$$\therefore L_1 = \left(\frac{12}{12+4+3} \times 114 \right) = 72\text{cm}$$

78. Key: (D)

Sol: Frequency of 2nd harmonic of open pipe,

$$f_1 = \frac{v}{\lambda} = \frac{v}{\ell} \quad \dots (i)$$

Frequency of nth harmonic of closed pipe,

$$f_2 = \frac{v}{\lambda} = \frac{nv}{4\ell} \quad \dots (ii)$$

Here n is a odd number. From eq. (i) and (ii)

$$f_2 = \frac{n}{4} f_1, \text{ For first resonance, } n = 5$$

$$\therefore f_2 = \frac{5}{4} f_1$$

79. Key: 5

Sol: The distance between two successive nodes

$$D = \frac{\lambda}{2} = \frac{v}{2f} = \frac{\sqrt{T/\mu}}{2f} = \frac{\sqrt{\frac{0.5 \times 0.2}{10^{-3}}}}{2 \times 100} = \frac{10}{2} = 5\text{cm}$$

80. Key: 35.00

Sol: Given, Density of wire, $\sigma = 9 \times 10^{-3} \text{kg cm}^{-3}$

Young's modulus of wire, $Y = 9 \times 10^{10} \text{Nm}^{-2}$

$$\text{Strain} = 4.9 \times 10^{-4}$$

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{T/A}{\text{Strain}}$$

$$\therefore \frac{T}{A} = Y \times \text{Strain} = 9 \times 10^9 \times 4.9 \times 10^{-4}$$

Also, mass of wire, $m = A / \sigma$

$$\text{Mass per unit length, } \mu = \frac{m}{J} = A\sigma$$

Fundamental frequency in the string

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}} = \frac{1}{2l} \sqrt{\frac{T}{\sigma A}}$$

$$= \frac{1}{2 \times 1} \sqrt{\frac{9 \times 10^9 \times 4.9 \times 10^{-4}}{9 \times 10^3}} = \frac{1}{2} \sqrt{49 \times 10^{9-4-3}} = \frac{1}{2} \times 70 = 35\text{Hz}$$

81. Key: ACD

Sol: There should be a displacement node at $x = 0$ and a displacement antinode at $x = 3$ m. Therefore, $y = 0$ and $x = 0$ and $y = \pm A$ at $x = 3$ m.

$$\text{Speed of wave, } v = \frac{\omega}{k} = 100 \text{ ms}^{-1}$$

Hence options (a), (c) & (d) satisfy the above conditions.

82. Key: B

Sol: As $v = \frac{p}{2\ell} \sqrt{\frac{T}{m}} \therefore T = \frac{v^2 \ell^2 m}{p^2}$

$$\text{String - 1 } T_0 = \frac{f_0^2 4L_0^2 \mu}{\ell^2}$$

$$\text{String - 2 } T_2 = \frac{f_0^2 4 \left(\frac{3}{2}\right)^2 L_0^2 (2\mu)}{(3)^2} = \frac{T_0}{2}$$

$$\text{String - 3 } T_3 = \frac{f_0^2 4 \left(\frac{5}{2}\right)^2 L_0^2 (3\mu)}{5^2} = \frac{3}{16} T_0$$

$$\text{String - 4 } T_4 = \frac{f_0^2 4 \left(\frac{7}{4}\right)^2 L_0^2 (4\mu)}{(14)^2} = \frac{T_0}{16}$$

83. Key: 336m/s

Sol: In the fundamental mode

$$(\ell + 0.6r) = \frac{\lambda}{4} = \frac{v}{4f} \Rightarrow v = 4f(\ell + 0.6r)$$

Here, $0.6r$ = end - correction in tube

r (= radius of tube) = 2.5 cm

ℓ (= length of tube) = 16 cm

f (= frequency of tuning fork) = 480 Hz

$\therefore$ Velocity of sound in air, $v = 336 \text{ m/s}$.

84. Key: C

Sol: Beat frequency

= difference in frequencies of two waves

$$= 11 - 9 = 2 \text{ Hz}$$

85. Key: A

Sol: $-r + \frac{v}{2 \times 0.95} = 5$

$$r - \frac{v}{2 \times 1} = 5$$

$$\text{Solving } r = 195 \text{ Hz}$$

$$v = 380 \text{ ms}^{-1}$$

86. Key: B

Sol: For interference to take place the wave should reflect back and come back to starting point where the second wave is starting

Distance travelled by first wave

$$= 2 \times 0.4 \text{ m}$$

$$= 0.8 \text{ m}$$

Velocity of wave

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{m/c}}$$

$$= \sqrt{1.6 \times 0.4 \times 10^{-2}} \text{ m/s}$$

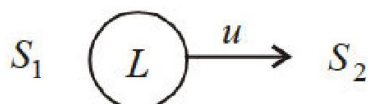
$$= 8 \text{ m/s}$$

$$\Delta T = \frac{\text{distance}}{\text{speed}} = \frac{0.8 \text{ m}}{8} = 0.10 \text{ sec}$$

87. Key: C

Sol: $f_1 = f \frac{v-v_0}{v}$ and $f_2 = f \frac{v+v_0}{v}$

But frequency,



$$f_2 - f_1 = f \times \frac{2v_0}{v}$$

$$\text{Or } 10 = 660 \times \frac{2u}{330}$$

$$\therefore u = 2.5 \text{ m/s}$$

88. Key: A

Sol: Apparent frequency of the siren heard by the car driver, $f' = f \left[\frac{v+v_0}{v-v_s} \right]$

Here $v = 320 \text{ m/s}$ (given)

$$v_0 = v_s = 36 \times \frac{5}{18} = 10 \text{ m/s}$$

$$\text{Or, } f' = 8 \left[\frac{320+10}{320-10} \right] = 8 \times \frac{33}{31} \approx 8.5 \text{ kHz}$$

89. Key: B

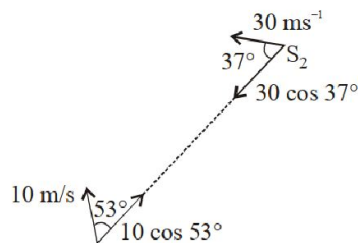
Sol: Using the formula $f' = f \left(\frac{v_A + v}{v} \right)$

$$\frac{v_A + v}{v} = \frac{5.5}{5} \text{ and } \frac{v_B + v}{v} = \frac{6}{5} \therefore \frac{v_B}{v_A} = 2$$

90. Key: 8.13

Sol: Apparent frequency heard by observer due to source S_1

$$v_1 = v \left[\frac{v+v_0}{v} \right] = 120 \left[\frac{330+10}{330} \right] = 120 \times \frac{34}{33} = 123.636 \text{ Hz}$$

Apparent frequency heard by observer due to source S_2

$$v_2 = v \left[\frac{v + v_0}{v - v_s} \right] = 120 \left[\frac{330 + 10 \cos 53^\circ}{330 - 30 \cos 37^\circ} \right]$$

$$\therefore v_2 = 120 \left[\frac{330 + 10 \times 0.6}{330 - 30 \times 0.8} \right] = 120 \left[\frac{336}{306} \right] = 131.764 \text{ Hz}$$

$$\therefore \text{Beat frequency, } v_b = v_2 - v_1 = 131.764 - 123.636 = 8.125 \text{ Hz}$$

$$\approx 8.13 \text{ Hz}$$

91. Key: 5

Sol: Apparent frequency at O due to source at - A

$$v_A = v \left[\frac{v}{v - 2 \cos \theta} \right]$$

Apparent frequency at 'O' due to source at - B

$$v_B = v \left[\frac{v}{v + 1 \cos \theta} \right]$$

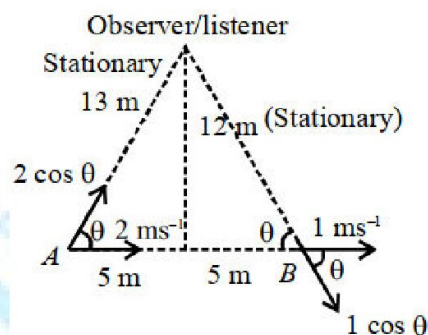
$$\therefore \text{Beat frequency, } v_b = v \left[\frac{v}{v - 2 \cos \theta} \right] - v \left[\frac{v}{v + \cos \theta} \right]$$

$$= v \left[\frac{1}{v - 2 \cos \theta} - \frac{1}{v + \cos \theta} \right]$$

$$= 1430 \times 330 \left[\frac{1}{330 - 2 \times \frac{5}{13}} - \frac{1}{330 + \frac{5}{13}} \right]$$

$$= 1430 \times 330 \times 13 \left[\frac{1}{330 \times 13 - 10} - \frac{1}{330 \times 13 + 5} \right]$$

$$= 1430 \times 330 \times 13 \left[\frac{1}{4280} - \frac{1}{4295} \right] \approx 5 \text{ Hz}$$



92. Key: 6

Sol: Frequency observed at car

$$v_1 = v \left(\frac{v + v_c}{v} \right)$$

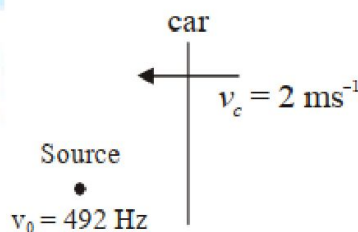
Frequency of reflected sound as observed at the source

$$v_2 = v_1 \left(\frac{v}{v - v_c} \right) = v_0 \left(\frac{v + v_c}{v - v_c} \right)$$

$$\therefore \text{Beat frequency} = v_2 - v_0$$

$$= v_0 \left[\frac{v + v_c - v + v_c}{v - v_c} \right] = v_0 \left[\frac{v + v_c - v + v_c}{v - v_c} \right]$$

$$= v_0 \left[\frac{2v_c}{v - v_c} \right] = 492 \left[\frac{2 \times 2}{330 - 2} \right] = \frac{492 \times 4}{328} = 6 \text{ Hz}$$



93. Key: (D)

Sol: $f = 244 \text{ Hz}$

$$\frac{\lambda}{4} = 0.356 \pm 0.005 \Rightarrow \lambda = 1.400 \pm 0.020$$

$$\Rightarrow v = (341.6 \pm 4.88) \frac{\text{m}}{\text{s}}$$

$$v = \lambda f = v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{100\gamma RT}{100M}}$$

$$\frac{\sqrt{100R\gamma T}}{\sqrt{100 \times M \times 10^{-3} \text{ kg/mol}}}$$

For neon : $640 \times \frac{7}{10} = 448 \text{ ms}^{-1}$

For Nitrogen: $590 \times \frac{3}{5} = 384 \text{ ms}^{-1}$

For Oxygen: $590 \times \frac{9}{16} = 351.875 \text{ ms}^{-1}$

For Argon: $640 \times \frac{17}{32} = 340 \text{ ms}^{-1}$

94. Key: (B)

Sol: The speed of each pulse is 2cm/s and initially two pulses are 8 cm apart and moving towards each other.



After two seconds pulses will overlap each other. According to superposition principle the string will not have any distortion and will be straight.

Hence there will be no P.E. The total energy will be only kinetic.

Passage – 3

Key: (B)

Sol: The speed of sound depends on the frame of reference of the observer.

$$V_{SA} = 340 + 20 = 360 \text{ m/s and } V_{SB} = 340 - 30 = 310 \text{ m/s}$$

96. Key: A

Sol: There is no relative motion between source and observer for the passengers in train A. since all the passengers in train A are moving with a velocity of 20m/s therefore the distribution of sound intensity of the whistle by the passengers in train A is uniform.

97. Key: (A)

Sol: For the passengers in train B, source is approaching with velocity 20m/s and observer is receding with velocity 30 m/s

$$v' = v_1 \left[\frac{v - v_0}{v - v_s} \right] = 800 \left[\frac{340 - 30}{340 - 20} \right] = 800 \times \frac{31}{32}$$

$$v'' = v_2 \left[\frac{v - v_0}{v - v_s} \right] = 1120 \times \frac{31}{32}$$

$$\therefore v'' - v' = (1120 - 800) \times \frac{31}{32} = 320 \times \frac{31}{32} = 310 \text{ Hz}$$